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Light-emitting device, light-emitting apparatus, electronic device, and lighting device

Abstract

Provided is a light-emitting device that can have sufficient device characteristics even with a smaller number of stacked layers in an EL layer and a thicker light-emitting layer than a conventional one by using, as an organic compound used for the EL layer of the light-emitting device, a material that can increase not only a property of carrier transport to the light-emitting layer but also a property of carrier injection from an electrode.

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Background/Summary

(1) This application is a 371 of international application PCT/IB2019/056350 filed on Jul. 25, 2019 which is incorporated herein by reference.

TECHNICAL FIELD

(2) One embodiment of the present invention relates to a light-emitting device, a light-emitting apparatus, an electronic device, and a lighting device. However, embodiments of the present invention are not limited thereto. That is, one embodiment of the present invention relates to an object, a method, a manufacturing method, or a driving method. Alternatively, one embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter.

BACKGROUND ART

(3) A light-emitting device including an EL layer between a pair of electrodes (also referred to as an organic EL device) has characteristics such as thinness, light weight, high-speed response to input signals, and low power consumption; thus, a display including such a light-emitting device has attracted attention as a next-generation flat panel display.

(4) In a light-emitting device, voltage application between a pair of electrodes causes, in an EL layer, recombination of electrons and holes injected from the electrodes, which brings a light-emitting substance (organic compound) contained in the EL layer into an excited state, and the light-emitting substance emits light when returning to the ground state from the excited state. The excited state can be a singlet excited state (S*) and a triplet excited state (T*). Light emission from a singlet excited state is referred to as fluorescence, and light emission from a triplet excited state is referred to as phosphorescence. The statistical generation ratio thereof in the light-emitting device is considered to be S*: T*=1:3. Since the emission spectrum obtained from a light-emitting

substance depends on the light-emitting substance, the use of different types of organic compounds as light-emitting substances makes it possible to obtain light-emitting devices which exhibit various emission colors.

(5) In order to improve device characteristics of such a light-emitting device, improvement of a device structure, development of a material, and the like have been actively carried out (see Patent Document 1, for example).

REFERENCE

Patent Document

(6) [Patent Document 1] Japanese Published Patent Application No. 2010-182699

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(7) In order to improve device characteristics of a light-emitting device, it is desirable that carriers be efficiently recombined at a desired position in a light-emitting layer of an EL layer.

(8) This needs an improvement in the property of injecting and transporting carriers from both electrodes. In addition, in order to increase the reliability of the device, it is necessary to increase carriers existing in the light-emitting layer and to increase the tolerance of the device to damage caused by driving. Thus, in one embodiment of the present invention, a novel light-emitting device whose device characteristics can be improved without relying on stacking layers alone is provided. Furthermore, a novel light-emitting device whose reliability can be improved is provided.

(9) Note that the descriptions of these objects do not preclude the existence of other objects. One embodiment of the present invention does not have to achieve all these objects. Other objects will be apparent from the descriptions of the specification, the drawings, the claims, and the like, and other objects can be derived from the descriptions of the specification, the drawings, the claims, and the like.

Means for Solving the Problems

(10) One embodiment of the present invention is a light-emitting device that can have sufficient device characteristics even with a smaller number of stacked layers in an EL layer and a thicker light-emitting layer than a conventional one by using, as an organic compound used for the EL layer of the light-emitting device, a material that can increase not only a property of carrier transport to the light-emitting layer but also a property of carrier injection from an electrode.

(11) One embodiment of the present invention is a light-emitting device including an EL layer between an anode and a cathode, in which the EL layer includes a first layer and a second layer; the first layer is in contact with the second layer and positioned between the anode and the second layer; the second layer includes a light-emitting substance, a first organic compound, and a second organic compound; the first organic compound has a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton; and the first layer includes the second organic compound.

(12) In the light-emitting device having the above structure, the EL layer includes the first layer, the second layer, and a third layer. The third layer is in contact with the second layer and positioned between the cathode and the second layer. The third layer includes a third organic compound, and the third organic compound is different from the first organic compound.

(13) In the above structure, the light-emitting device includes a fifth layer in contact with the third layer, and the fifth layer includes a compound containing an alkali metal.

(14) In the above structure, the EL layer includes the first layer, the second layer, and a fourth layer. The fourth layer is in contact with the cathode, and the fourth layer includes the first organic compound and a compound containing an alkali metal.

(15) Another embodiment of the present invention is a light-emitting device including an EL layer between an anode and a cathode, in which the EL layer includes a second layer and a fourth layer; the fourth layer is in contact with the cathode; the second layer contains a light-emitting substance; and the fourth layer includes a first organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton and a compound containing an alkali metal.

(16) In the above structure, the fourth layer is in contact with the second layer.

(17) In any one of the above structures, the first organic compound has a structure in which an aromatic ring is fused to a furan ring of a furopyrazine skeleton or a fuopyrimidine skeleton.

(18) Any one of the above structures, the first organic compound is represented by any one of General Formula (G1) below, General Formula (G2) below, and General Formula (G3) below.

(19) ##STR00001##

(20) (In the formulae, Ar.sup.1 represents a substituted or unsubstituted aromatic ring.

(21) Furthermore, each of R.sup.1 to R.sup.6 independently represents hydrogen or a group having 1 to 100 carbon atoms in total, and at least one of R.sup.1 and R.sup.2, at least one of R.sup.3 and R.sup.4, and at least one of R.sup.5 and R.sup.6 each have a hole-transport skeleton.

(22) Any one of the above structures, the first organic compound is represented by any one of General Formula (G1) below, General Formula (G2) below, and General Formula (G3) below.

(23) ##STR00002##

(24) (In the formulae, Ar.sup.1 represents any one of substituted or unsubstituted benzene, substituted or unsubstituted naphthalene, substituted or unsubstituted phenanthrene, and substituted or unsubstituted chrysene. Furthermore, each of R.sup.1 to R.sup.6 independently represents hydrogen or a group having 1 to 100 carbon atoms in total, and at least one of R.sup.1 and R.sup.2, at least one of R.sup.3 and R.sup.4, and at least one of R.sup.5 and R.sup.6 each have a hole-transport skeleton.

(25) In the above two structures, Ar.sup.1 in General Formula (G1), General Formula (G2), or General Formula (G3) is one of General Formula (t1) to General Formula (t4) below.

(26) ##STR00003##

(27) (In the formulae, each of R.sup.11 to R.sup.36 independently represents any one of hydrogen, a substituted or unsubstituted alkyl group having 1 to 6 carbon atoms, a substituted or unsubstituted monocyclic saturated hydrocarbon group having 3 to 7 carbon atoms, a substituted or unsubstituted aromatic hydrocarbon group having 6 to 30 carbon atoms, and a substituted or unsubstituted heteroaromatic hydrocarbon group having 3 to 12 carbon atoms. Furthermore, * represents a portion bonded to a furan ring of a furopyrazine skeleton or a fuopyrimidine skeleton in General Formula (G1) to General Formula (G3).)

(28) In the above plurality of structures, each of R.sup.1 to R.sup.6, the group having 1 to 100 carbon atoms in total in General Formula (G1) to General Formula (G3), represents one of or a combination of an alkyl group having 1 to 6 carbon atoms, a substituted or unsubstituted monocyclic saturated hydrocarbon group having 5 to 7 carbon atoms, a substituted or unsubstituted polycyclic saturated hydrocarbon group having 7 to 10 carbon atoms, a substituted or unsubstituted aromatic hydrocarbon group having 6 to 13 carbon atoms, and a substituted or unsubstituted heteroaromatic hydrocarbon group having 3 to 12 carbon atoms.

(29) In the above plurality of structures, each of R.sup.1 to R.sup.6, the group having 1 to 100 carbon atoms in total in General Formula (G1) to General Formula (G3), has any one of a pyrrole ring structure, a furan ring structure, and a thiophene ring structure through a substituted or unsubstituted phenylene group or a substituted or unsubstituted biphenylene group.

(30) In the above plurality of structures, each of R.sup.1 to R.sup.6, the group having 1 to 100 carbon atoms in total in General Formula (G1) to General Formula (G3), has a structure represented by any one of General Formulae (Ht-1) to (Ht-26) below through a substituted or unsubstituted phenylene group or a substituted or unsubstituted biphenylene group.

(31) ##STR00004## ##STR00005## ##STR00006## ##STR00007## ##STR00008##

(32) (In the formulae, Q represents oxygen or sulfur. Each of R.sup.100 to R.sup.169 represents 1 to 4 substituents and independently represents any one of hydrogen, an alkyl group having 1 to 6 carbon atoms, and a substituted or unsubstituted aromatic hydrocarbon group having 6 to 13 carbon atoms. In addition, Ar.sup.1 represents a substituted or unsubstituted aromatic hydrocarbon group having 6 to 13 carbon atoms.)

(33) The present invention also includes a light-emitting device having any of the above structures, in which the light-emitting substance is a phosphorescent material. Furthermore, the present invention also includes a light-emitting device including such a light-emitting substance and the second organic compound that is a carbazole derivative, preferably a bicarbazole derivative.

(34) Note that one embodiment of the present invention includes, in its category, in addition to a light-emitting apparatus including the above-described light-emitting device, an electronic device including a light-emitting device or a light-emitting apparatus (specifically, an electronic device including a light-emitting device or a light-emitting apparatus and a connection terminal or an operation key) and a lighting device including a light-emitting device or a light-emitting apparatus (specifically, a lighting device including a light-emitting device or a light-emitting apparatus and a housing). Accordingly, a light-emitting apparatus in this specification refers to an image display device or a light source (including a lighting apparatus). In addition, a light-emitting apparatus includes a module in which a light-emitting apparatus is connected to a connector such as an FPC (Flexible Printed Circuit) or a TCP (Tape Carrier Package), a module in which a printed wiring board is provided on the tip of a TCP, or a module in which an IC (integrated circuit) is directly mounted on a light-emitting device by a COG (Chip On Glass) method.

Effect of the Invention

(35) According to one embodiment of the present invention, a novel light-emitting device whose device characteristics can be improved can be provided. According to another embodiment of the present invention, a novel light-emitting device whose device characteristics can be improved without relying on stacking layers alone can be provided.

(36) Note that the descriptions of the effects do not preclude the existence of other effects. One embodiment of the present invention does not need to have all these effects. Other effects will be apparent from the descriptions of the specification, the drawings, the claims, and the like, and other effects can be derived from the descriptions of the specification, the drawings, the claims, and the like. In addition, a novel light-emitting device whose reliability can be improved can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A and 1B are diagrams each illustrating a structure of a light-emitting device.

(2) FIGS. 2A-2C are diagrams illustrating light-emitting apparatuses.

(3) FIGS. 3A and 3B are diagrams illustrating a light-emitting apparatus.

(4) FIGS. 4A-4G are diagrams each illustrating an electronic device.

(5) FIGS. 5A-5C are diagrams illustrating an electronic device.

(6) FIGS. 6A and 6B are diagrams illustrating an automobile.

(7) FIGS. 7A and 7B are diagrams each illustrating a lighting device.

(8) FIG. 8 is a diagram illustrating a light-emitting device.

(9) FIG. 9 is a diagram showing luminance-current density characteristics of a light-emitting device 1, a light-emitting device 2, and a comparative light-emitting device 3.

(10) FIG. 10 is a diagram showing luminance-voltage characteristics of the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3.

(11) FIG. 11 is a diagram showing current efficiency-luminance characteristics of the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3.

(12) FIG. 12 is a diagram showing current-voltage characteristics of the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3.

(13) FIG. 13 is a diagram showing emission spectra of the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3.

(14) FIG. 14 is a diagram showing reliability of the light-emitting device 1, the light-emitting

device 2, and the comparative light-emitting device 3.

MODE FOR CARRYING OUT THE INVENTION

(15) Embodiments of the present invention are described in detail below with reference to drawings. Note that the present invention is not limited to the following description, and the modes and details of the present invention can be modified in various ways without departing from the spirit and scope of the present invention. Thus, the present invention should not be construed as being limited to the descriptions in the following embodiments.

(16) Note that the position, size, range, or the like of each structure illustrated in the drawings and the like do not represent the actual position, size, range, or the like in some cases for easy understanding. Therefore, the disclosed invention is not necessarily limited to the position, size, range, or the like disclosed in drawings and the like.

(17) Furthermore, when describing the structures of the invention with reference to the drawings in this specification and the like, the reference numerals denoting the same components are commonly used in different drawings.

Embodiment 1

(18) In this embodiment, a light-emitting device of one embodiment of the present invention is described with reference to FIG. 1.

(19) <<Structure of Light-Emitting Device>>

(20) FIG. 1 illustrates examples of a light-emitting device including, between a pair of electrodes, an EL layer having a light-emitting layer. Specifically, the light-emitting device has a structure in which an EL layer **103** is sandwiched between a first electrode **101** and a second electrode **102**. Note that the EL layer **103** has a structure in which, for example, a hole-injection layer **111**, a hole-transport layer **112**, a light-emitting layer **113**, an electron-transport layer **114**, and an electron-injection layer **115** are sequentially stacked as functional layers, in the case where the first electrode **101** serves as an anode. Embodiments of the present invention also include light-emitting devices having other structures: for example, a light-emitting device that can be driven at a low voltage by having a structure (a tandem structure) in which a plurality of EL layers, between which a charge-generation layer is sandwiched, are provided between a pair of electrodes; and a light-emitting device that has improved optical characteristics by having a micro-optical resonator (microcavity) structure between a pair of electrodes. Note that the charge generation layer has a function of injecting electrons into one of the adjacent EL layers and injecting holes into the other of the EL layers when a voltage is applied to the first electrode **101** and the second electrode **102**.

(21) Note that at least one of the first electrode **101** and the second electrode **102** of the above light-emitting device is an electrode having a light-transmitting property (e.g., a transparent electrode or a semi-transmissive and semi-reflective electrode). In the case where the electrode having a light-transmitting property is a transparent electrode, the visible light transmittance of the transparent electrode is 40% or higher. In the case where the electrode having a light-transmitting property is a semi-transmissive and semi-reflective electrode, the visible light reflectance of the semi-transmissive and semi-reflective electrode is higher than or equal to 20% and lower than or equal to 80%, preferably higher than or equal to 40% and lower than or equal to 70%. The resistivity of these electrodes is preferably $1 \times 10^{-2} \Omega\text{cm}$ or lower.

(22) Furthermore, when one of the first electrode **101** and the second electrode **102** is an electrode having reflectivity (reflective electrode) in the above light-emitting device of one embodiment of the present invention, the visible light reflectance of the reflective electrode is higher than or equal to 40% and lower than or equal to 100%, preferably higher than or equal to 70% and lower than or equal to 100%. The resistivity of this electrode is preferably $1 \times 10^{-2} \Omega\text{cm}$ or lower.

(23) <First Electrode and Second Electrode>

(24) As materials for forming the first electrode **101** and the second electrode **102**, any of the following materials can be used in an appropriate combination as long as the functions of the electrodes described above can be fulfilled. For example, a metal, an alloy, an electrically

conductive compound, a mixture of these, and the like can be used as appropriate. Specifically, an In—Sn oxide (also referred to as ITO), an In—Si—Sn oxide (also referred to as ITSO), an In—Zn oxide, or an In—W—Zn oxide can be given. In addition, it is also possible to use a metal such as aluminum (Al), titanium (Ti), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni), copper (Cu), gallium (Ga), zinc (Zn), indium (In), tin (Sn), molybdenum (Mo), tantalum (Ta), tungsten (W), palladium (Pd), gold (Au), platinum (Pt), silver (Ag), yttrium (Y), or neodymium (Nd) or an alloy containing an appropriate combination of any of these metals. It is also possible to use an element belonging to Group 1 or Group 2 in the periodic table, which is not listed above as an example (for example, lithium (Li), cesium (Cs), calcium (Ca), or strontium (Sr)), a rare earth metal such as europium (Eu) or ytterbium (Yb), an alloy containing an appropriate combination of any of these elements, graphene, or the like.

(25) For fabrication of these electrodes, a sputtering method or a vacuum evaporation method can be used.

(26) <Hole-Injection Layer>

(27) The hole-injection layer **111** is a layer injecting holes from the first electrode **101** that is an anode to the EL layer **103**, and is a layer containing an organic acceptor material or a material with a high hole-injection property.

(28) The organic acceptor material is a material that allows holes to be generated in another organic compound whose HOMO level value is close to the LUMO level value of the organic acceptor material when charge separation is caused between the organic acceptor material and the organic compound. Thus, as the organic acceptor material, a compound having an electron-withdrawing group (a halogen group or a cyano group), such as a quinodimethane derivative, a chloranil derivative, and a hexaazatriphenylene derivative, can be used. For example, 7,7,8,8-tetracyano-2,3,5,6-tetrafluoroquinodimethane (abbreviation: F4-TCNQ), 3,6-difluoro-2,5,7,7,8,8-hexacyanoquinodimethane, chloranil, 2,3,6,7,10,11-hexacyano-1,4,5,8,9,12-hexaazatriphenylene (abbreviation: HAT-CN), or 1,3,4,5,7,8-hexafluorotetracyano-naphthoquinodimethane (abbreviation: F6-TCNNQ) can be used. Among organic acceptor materials, HAT-CN, which has a high acceptor property and stable film quality against heat, is particularly favorable. Besides, a [3]radialene derivative has a very high electron-accepting property and thus is preferable; specifically, α,α',α'' -1,2,3-cyclopropanetriylidenetris[4-cyano-2,3,5,6-tetrafluorobenzeneacetonitrile], α,α',α'' -1,2,3-cyclopropanetriylidenetris[2,6-dichloro-3,5-difluoro-4-(trifluoromethyl)benzeneacetonitrile], α,α',α'' -1,2,3-cyclopropanetriylidenetris[2,3,4,5,6-pentafluorobenzeneacetonitrile], or the like can be used.

(29) Examples of the material with a high hole-injection property include transition metal oxides such as molybdenum oxide, vanadium oxide, ruthenium oxide, tungsten oxide, and manganese oxide. Alternatively, it is possible to use a phthalocyanine-based compound such as phthalocyanine (abbreviation: H.sub.2Pc) or copper phthalocyanine (abbreviation: CuPc), or the like.

(30) In addition to the above materials, it is also possible to use an aromatic amine compound, which is a low molecular compound, such as 4,4',4''-tris(N,N-diphenylamino)triphenylamine (abbreviation: TDATA), 4,4',4''-tris[N-(3-methylphenyl)-N-phenylamino]triphenylamine (abbreviation: MTDATA), 4,4'-bis[N-(4-diphenylaminophenyl)-N-phenylamino]biphenyl (abbreviation: DPAB), 4,4'-bis[N-{4-[N-(3-methylphenyl)-N-phenylamino]phenyl}-N-phenylamino]biphenyl (abbreviation: DNTPD), 1,3,5-tris[N-(4-diphenylaminophenyl)-N-phenylamino]benzene (abbreviation: DPA3B), 3-[N-(9-phenylcarbazol-3-yl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzPCA1), 3,6-bis[N-(9-phenylcarbazol-3-yl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzPCA2), or 3-[N-(1-naphthyl)-N-(9-phenylcarbazol-3-yl)amino]-9-phenylcarbazole (abbreviation: PCzPCN1).

(31) It is also possible to use a high molecular compound (an oligomer, a dendrimer, a polymer, or the like) such as poly(N-vinylcarbazole) (abbreviation: PVK), poly(4-vinyltriphenylamine) (abbreviation: PVTPA), poly[N-(4-{N-[4-(4-diphenylamino)phenyl]phenyl}-N-

phenylamino}phenyl)methacrylamide](abbreviation: PTPDMA), or poly[N,N-bis(4-butylphenyl)-N,N-bis(phenyl)benzidine](abbreviation: Poly-TPD). Alternatively, it is also possible to use a high molecular compound to which acid is added, such as poly(3,4-ethylenedioxythiophene)/poly(styrenesulfonic acid) (abbreviation: PEDOT/PSS) or polyaniline/poly(styrenesulfonic acid) (abbreviation: PAni/PSS).

(32) Alternatively, as the material having a high hole-injection property, a composite material containing a hole-transport material and an acceptor material (electron-accepting material) can be used. In this case, the acceptor material extracts electrons from a hole-transport material, so that holes are generated in the hole-injection layer **111** and the holes are injected into the light-emitting layer **113** through the hole-transport layer **112**. Note that the hole-injection layer **111** may be formed to have a single-layer structure of a composite material containing a hole-transport material and an acceptor material (electron-accepting material), or a stacked-layer structure in which a layer containing a hole-transport material and a layer containing an acceptor material (electron-accepting material) are stacked.

(33) As the hole-transport material, a substance having a hole mobility of greater than or equal to 1×10^{-6} cm²/Vs is preferable. Note that other substances can be used as long as they have a property of transporting more holes than electrons.

(34) As the hole-transport material, materials having a high hole-transport property, such as a π -electron rich heteroaromatic compound (e.g., a carbazole derivative and a furan derivative) and an aromatic amine (a compound having an aromatic amine skeleton), are preferable.

(35) Examples of the above carbazole derivative (a compound having a carbazole skeleton) include a bicarbazole derivative (e.g., a 3,3'-bicarbazole derivative) and an aromatic amine having a carbazolyl group.

(36) Note that specific examples of the bicarbazole derivative (e.g., a 3,3'-bicarbazole derivative) include 3,3'-bis(9-phenyl-9H-carbazole) (abbreviation: PCCP), 9,9'-bis(1,1'-biphenyl-4-yl)-3,3'-bi-9H-carbazole, 9,9'-bis(1,1'-biphenyl-3-yl)-3,3'-bi-9H-carbazole, 9-(1,1'-biphenyl-3-yl)-9'-(1,1'-biphenyl-4-yl)-9H,9'H-3,3'-bicarbazole (abbreviation: mBPCCBP), 9-(2-naphthyl)-9'-phenyl-9H,9'H-3,3'-bicarbazole (abbreviation: pNCCP).

(37) Specific examples of the aromatic amine having the above carbazolyl group include 4-phenyl-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBA1BP), N-(4-biphenyl)-N-(9,9-dimethyl-9H-fluoren-2-yl)-9-phenyl-9H-carbazol-3-amine (abbreviation: PCBiF), N-(1,1'-biphenyl-4-yl)-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9-dimethyl-9H-fluoren-2-amine (abbreviation: PCBBiF), 4,4'-diphenyl-4''-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBBi1BP), 4-(1-naphthyl)-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBANB), 4,4'-di(1-naphthyl)-4''-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBNBB), 4-phenyldiphenyl-(9-phenyl-9H-carbazol-3-yl)amine (abbreviation: PCA1BP), N,N-bis(9-phenylcarbazol-3-yl)-N,N-diphenylbenzene-1,3-diamine (abbreviation: PCA2B), N,N,N'-triphenyl-N,N,N'-tris(9-phenylcarbazol-3-yl)benzene-1,3,5-triamine (abbreviation: PCA3B), 9,9-dimethyl-N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]fluoren-2-amine (abbreviation: PCBAF), N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]spiro-9,9'-bifluoren-2-amine (abbreviation: PCBASF), 3-[N-(9-phenylcarbazol-3-yl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzPCA1), 3,6-bis[N-(9-phenylcarbazol-3-yl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzPCA2), 3-[N-(1-naphthyl)-N-(9-phenylcarbazol-3-yl)amino]-9-phenylcarbazole (abbreviation: PCzPCN1), 3-[N-(4-diphenylaminophenyl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzDPA1), 3,6-bis[N-(4-diphenylaminophenyl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzDPA2), 3,6-bis[N-(4-diphenylaminophenyl)-N-(1-naphthyl)amino]-9-phenylcarbazole (abbreviation: PCzTPN2), 2-[N-(9-phenylcarbazol-3-yl)-N-phenylamino]spiro-9,9'-bifluorene (abbreviation: PCASF), N-[4-(9H-carbazol-9-yl)phenyl]-N-(4-phenyl)phenylaniline (abbreviation: YGA1BP), N,N-bis[4-(carbazol-9-yl)phenyl]-N,N-diphenyl-9,9-dimethylfluorene-2,7-diamine (abbreviation: YGA2F), and 4,4',4''-tris(carbazol-9-

yl)triphenylamine (abbreviation: TCTA).

(38) In addition to the above, other examples of the carbazole derivative include 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn), 3-[4-(1-naphthyl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPN), 1,3-bis(N-carbazolyl)benzene (abbreviation: mCP), 4,4'-di(N-carbazolyl)biphenyl (abbreviation: CBP), 3,6-bis(3,5-diphenylphenyl)-9-phenylcarbazole (abbreviation: CzTP), 1,3,5-tris[4-(N-carbazolyl)phenyl]benzene (abbreviation: TCPB), and 9-[4-(10-phenyl-9-anthracenyl)phenyl]-9H-carbazole (abbreviation: CzPA).

(39) Specific examples of the above furan derivative (a compound having a furan skeleton) include compounds having a thiophene skeleton, such as 4,4',4''-(benzene-1,3,5-triyl)tri(dibenzothiophene) (abbreviation: DBT3P-II), 2,8-diphenyl-4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]dibenzothiophene (abbreviation: DBTFLP-III), and 4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]-6-phenyldibenzothiophene (abbreviation: DBTFLP-IV), and compounds having a furan skeleton, such as 4,4',4''-(benzene-1,3,5-triyl)tri(dibenzofuran) (abbreviation: DBF3P-II) and 4-{3-[3-(9-phenyl-9H-fluoren-9-yl)phenyl]phenyl}dibenzofuran (abbreviation: mmDBFFLBI-II).

(40) Specific examples of the above aromatic amine include 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (abbreviation: NPB or α -NPD), N,N-bis(3-methylphenyl)-N,N-diphenyl-[1,1'-biphenyl]-4,4'-diamine (abbreviation: TPD), 4,4'-bis[N-(spiro-9,9'-bifluoren-2-yl)-N-phenylamino]biphenyl (abbreviation: BSPB), 4-phenyl-4'-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: BPAFLP), 4-phenyl-3'-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: mBPAFLP), N-(9,9-dimethyl-9H-fluoren-2-yl)-N-{9,9-dimethyl-2-[N-phenyl-N-(9,9-dimethyl-9H-fluoren-2-yl)amino]-9H-fluoren-7-yl}phenylamine (abbreviation: DFLADFL), N-(9,9-dimethyl-2-diphenylamino-9H-fluoren-7-yl)diphenylamine (abbreviation: DPNF), 2-[N-(4-diphenylaminophenyl)-N-phenylamino]spiro-9,9'-bifluorene (abbreviation: DPASF), 2,7-bis[N-(4-diphenylaminophenyl)-N-phenylamino]spiro-9,9'-bifluorene (abbreviation: DPA2SF), 4,4',4''-tris[N-(1-naphthyl)-N-phenylamino]triphenylamine (abbreviation: 1'-TNATA), 4,4',4''-tris(N,N-diphenylamino)triphenylamine (abbreviation: TDATA), 4,4',4''-tris[N-(3-methylphenyl)-N-phenylamino]triphenylamine (abbreviation: m-MTDATA), N,N'-di(p-tolyl)-N,N-diphenyl-p-phenylenediamine (abbreviation: DTDPPA), 4,4'-bis[N-(4-diphenylaminophenyl)-N-phenylamino]biphenyl (abbreviation: DPAB), N,N-bis{4-[bis(3-methylphenyl)amino]phenyl}-N,N'-diphenyl-(1,1'-biphenyl)-4,4'-diamine (abbreviation: DNTPD), and 1,3,5-tris[N-(4-diphenylaminophenyl)-N-phenylamino]benzene (abbreviation: DPA3B).

(41) As the hole-transport material, a high molecular compound such as poly(N-vinylcarbazole) (abbreviation: PVK), poly(4-vinyltriphenylamine) (abbreviation: PVTTPA), poly[N-(4-{N'-[4-(4-diphenylamino)phenyl]phenyl-N-phenylamino}phenyl)methacrylamide] (abbreviation: PTPDMA), or poly[N,N-bis(4-butylphenyl)-N,N-bis(phenyl)benzidine] (abbreviation: Poly-TPD) can also be used.

(42) Note that the hole-transport material is not limited to the above, and one of or a combination of various known materials may be used as the hole-transport material.

(43) As the acceptor material used for the hole-injection layer **111**, an oxide of a metal belonging to any of Group 4 to Group 8 of the periodic table can be used. As specific examples, molybdenum oxide, vanadium oxide, niobium oxide, tantalum oxide, chromium oxide, tungsten oxide, manganese oxide, and rhenium oxide can be given. Among these, molybdenum oxide is particularly preferable since it is stable in the air, has a low hygroscopic property, and is easy to handle. It is also possible to use any of the above-described organic acceptor materials.

(44) Note that the hole-injection layer **111** can be formed by any of various known deposition methods, and can be formed by a vacuum evaporation method, for example.

(45) <Hole-Transport Layer>

(46) The hole-transport layer **112** is a layer transporting holes, which are injected from the first electrode **101** through the hole-injection layer **111**, to the light-emitting layer **113**. Note that the hole-transport layer **112** is a layer containing a hole-transport material. Thus, for the hole-transport

layer **112**, a hole-transport material that can be used for the hole-injection layer **111** can be used.

(47) Note that in the light-emitting device of one embodiment of the present invention, the same organic compound as that for the hole-transport layer **112** is preferably used for the light-emitting layer **113**. This is because the use of the same organic compounds for the hole-transport layer **112** and the light-emitting layer **113** allows efficient hole transport from the hole-transport layer **112** to the light-emitting layer **113**.

(48) <Light-Emitting Layer>

(49) The light-emitting layer **113** is a layer containing a light-emitting substance. There is no particular limitation on the light-emitting substance that can be used for the light-emitting layer **113**, and it is possible to use a light-emitting substance that converts singlet excitation energy into light in the visible light range or a light-emitting substance that converts triplet excitation energy into light in the visible light range. In addition, a substance that exhibits emission color of blue, purple, bluish purple, green, yellowish green, yellow, orange, red, or the like can be appropriately used.

(50) In the light-emitting device of one embodiment of the present invention, the light-emitting layer **113** includes a light-emitting substance (guest material) and one or more kinds of organic compounds (e.g., host material). Note that as the organic compound (e.g., host material) used here, it is preferable to use a substance whose energy gap is larger than the energy gap of the light-emitting substance (guest material). Examples of one or more kinds of organic compounds (e.g., host material) include organic compounds such as a hole-transport material that can be used for the hole-transport layer **112** described above and an electron-transport material that can be used for the electron-transport layer **114** described later.

(51) In the case where a phosphorescent material is used as the light-emitting substance (guest material) in the light-emitting layer **113**, an organic compound such as a carbazole derivative (e.g., a bicarbazole derivative or an aromatic amine having a carbazolyl group) can be given as a material that is favorably combined with the light-emitting substance, among the above hole-transport materials. Furthermore, among the above electron-transport materials, an organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton can be given as a material that is favorably combined with the light-emitting substance, for example.

(52) As another structure, the light-emitting layer **113** may have a structure including a plurality of light-emitting layers containing different light-emitting substances to exhibit different emission colors (for example, white light emission obtained by a combination of complementary emission colors). Alternatively, a structure may be employed in which one light-emitting layer includes a plurality of different light-emitting substances.

(53) Examples of the light-emitting substance are given below.

(54) As an example of the light-emitting substance that converts singlet excitation energy into light emission, a substance that emits fluorescence (fluorescent material) can be given; examples include a pyrene derivative, an anthracene derivative, a triphenylene derivative, a fluorene derivative, a carbazole derivative, a dibenzothiophene derivative, a dibenzofuran derivative, a dibenzoquinoxaline derivative, a quinoxaline derivative, a pyridine derivative, a pyrimidine derivative, a phenanthrene derivative, and a naphthalene derivative. A pyrene derivative is particularly preferable because it has a high emission quantum yield. Specific examples of pyrene derivatives include N,N-bis(3-methylphenyl)-N,N-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (abbreviation: 1,6mMemFLPAPrn), N,N-diphenyl-N,N-bis[4-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (abbreviation: 1,6FLPAPrn), N,N-bis(dibenzofuran-2-yl)-N,N-diphenylpyrene-1,6-diamine (abbreviation: 1,6FrAPrn), N,N-bis(dibenzothiophen-2-yl)-N,N-diphenylpyrene-1,6-diamine (abbreviation: 1,6ThAPrn), N,N-(pyrene-1,6-diyl)bis[(N-phenylbenzo[b]naphtho[1,2-d]furan)-6-amine](abbreviation: 1,6BnfAPrn), N,N-(pyrene-1,6-diyl)bis[(N-phenylbenzo[b]naphtho[1,2-d]furan)-8-amine](abbreviation: 1,6BnfAPrn-02), and N,N-(pyrene-1,6-diyl)bis[(6,N-diphenylbenzo[b]naphtho[1,2-d]furan)-8-amine](abbreviation:

1,6BnFAPrn-03).

(55) In addition, it is possible to use 5,6-bis[4-(10-phenyl-9-anthryl)phenyl]-2,2'-bipyridine (abbreviation: PAP2BPy), 5,6-bis[4'-(10-phenyl-9-anthryl)biphenyl-4-yl]-2,2'-bipyridine (abbreviation: PAPP2BPy), N,N-bis[4-(9H-carbazol-9-yl)phenyl]-N,N-diphenylstilbene-4,4'-diamine (abbreviation: YGA2S), 4-(9H-carbazol-9-yl)-4'-(10-phenyl-9-anthryl)triphenylamine (abbreviation: YGAPA), 4-(9H-carbazol-9-yl)-4'-(9,10-diphenyl-2-anthryl)triphenylamine (abbreviation: 2YGAPPA), N,9-diphenyl-N-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazol-3-amine (abbreviation: PCAPA), 4-(10-phenyl-9-anthryl)-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBAPA), 4-[4-(10-phenyl-9-anthryl)phenyl]-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBAPBA), perylene, 2,5,8,11-tetra-tert-butylperylene (abbreviation: TBP), N,N'-(2-tert-butylanthracene-9,10-diyl-di-4,1-phenylene)bis[N,N,N-triphenyl-1,4-phenylenediamine] (abbreviation: DPABPA), N,9-diphenyl-N-[4-(9,10-diphenyl-2-anthryl)phenyl]-9H-carbazol-3-amine (abbreviation: 2PCAPPA), N-[4-(9,10-diphenyl-2-anthryl)phenyl]-N,N,N-triphenyl-1,4-phenylenediamine (abbreviation: 2DPAPPA), or the like.

(56) Examples of the light-emitting substance that converts triplet excitation energy into light emission include a substance that emits phosphorescence (phosphorescent material) and a thermally activated delayed fluorescence (TADF) material that exhibits thermally activated delayed fluorescence.

(57) Examples of a phosphorescent material include an organometallic complex, a metal complex (platinum complex), and a rare earth metal complex. These substances exhibit different emission colors (emission peaks), and thus are used through appropriate selection as needed.

(58) As examples of a phosphorescent material which exhibits blue or green and whose emission spectrum has a peak wavelength at greater than or equal to 450 nm and less than or equal to 570 nm, the following substances can be given.

(59) For example, organometallic complexes having a 4H-triazole skeleton, such as tris{2-[5-(2-methylphenyl)-4-(2,6-dimethylphenyl)-4H-1,2,4-triazol-3-yl-N.sup.2]phenyl-κC}iridium(III) (abbreviation: [Ir(mpptz-dmp).sub.3]), tris(5-methyl-3,4-diphenyl-4H-1,2,4-triazolato)iridium(III) (abbreviation: [Ir(Mptz).sub.3]), tris[4-(3-biphenyl)-5-isopropyl-3-phenyl-4H-1,2,4-triazolato]iridium(III) (abbreviation: [Ir(iPrptz-3b).sub.3]), and tris[3-(5-biphenyl)-5-isopropyl-4-phenyl-4H-1,2,4-triazolato]iridium(III) (abbreviation: [Ir(iPr5btz).sub.3]); organometallic complexes having a 1H-triazole skeleton, such as tris[3-methyl-1-(2-methylphenyl)-5-phenyl-1H-1,2,4-triazolato]iridium(III) (abbreviation: [Ir(Mptz1-mp).sub.3]) and tris(1-methyl-5-phenyl-3-propyl-1H-1,2,4-triazolato)iridium(III) (abbreviation: [Ir(Prptz1-Me).sub.3]); organometallic complexes having an imidazole skeleton, such as fac-tris[1-(2,6-diisopropylphenyl)-2-phenyl-1H-imidazole]iridium(III) (abbreviation: [Ir(iPrpmi).sub.3]) and tris[3-(2,6-dimethylphenyl)-7-methylimidazo[1,2-f]phenanthridinato]iridium(III) (abbreviation: [Ir(dmpimpt-Me).sub.3]); organometallic complexes in which a phenylpyridine derivative having an electron-withdrawing group is a ligand, such as bis[2-(4',6'-difluorophenyl)pyridinato-N,C.sup.2']iridium(III) tetrakis(1-pyrazolyl)borate (abbreviation: FIr6), bis[2-(4',6'-difluorophenyl)pyridinato-N,C.sup.2']iridium(III) picolinate (abbreviation: FIrpic), bis{2-[3',5'-bis(trifluoromethyl)phenyl]pyridinato-N,C.sup.2'}iridium(III) picolinate (abbreviation: [Ir(CF.sub.3ppy).sub.2(pic)]), and bis[2-(4',6'-difluorophenyl)pyridinato-N,C.sup.2']iridium(III) acetylacetonate (abbreviation: FIr(acac)); and the like can be given.

(60) As a phosphorescent material that exhibits green or yellow and whose emission spectrum has a peak wavelength at greater than or equal to 495 nm and less than or equal to 590 nm, the following substances can be given.

(61) For example, organometallic iridium complexes having a pyrimidine skeleton, such as tris(4-methyl-6-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(mppm).sub.3]), tris(4-tert-butyl-6-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(tBuppm).sub.3]), (acetylacetonato)bis(6-methyl-4-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(mppm).sub.2(acac)]), (acetylacetonato)bis(6-

tert-butyl-4-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(tBuppm).sub.2(acac)]), (acetylacetonato)bis[6-(2-norbornyl)-4-phenylpyrimidinato]iridium(III) (abbreviation: [Ir(nbppm).sub.2(acac)]), (acetylacetonato)bis[5-methyl-6-(2-methylphenyl)-4-phenylpyrimidinato]iridium(III) (abbreviation: [Ir(mppmm).sub.2(acac)]), (acetylacetonato)bis{4,6-dimethyl-2-[6-(2,6-dimethylphenyl)-4-pyrimidinyl-κN.sup.3]phenyl-κC}iridium(III) (abbreviation: [Ir(dmppm-dmp).sub.2(acac)]), and (acetylacetonato)bis(4,6-diphenylpyrimidinato)iridium(III) (abbreviation: [Ir(dppm).sub.2(acac)]); organometallic iridium complexes having a pyrazine skeleton, such as (acetylacetonato)bis(3,5-dimethyl-2-phenylpyrazinato)iridium(III) (abbreviation: [Ir(mppr-Me).sub.2(acac)]) and (acetylacetonato)bis(5-isopropyl-3-methyl-2-phenylpyrazinato)iridium(III) (abbreviation: [Ir(mppr-iPr).sub.2(acac)]); organometallic iridium complexes having a pyridine skeleton, such as tris(2-phenylpyridinato-N,C.sup.2')iridium(III) (abbreviation: [Ir(ppy).sub.3]), bis(2-phenylpyridinato-N,C.sup.2')iridium(III) acetylacetonate (abbreviation: [Ir(ppy).sub.2(acac)]), bis(benzo[h]quinolinato)iridium(III) acetylacetonate (abbreviation: [Ir(bzq).sub.2(acac)]), tris(benzo[h]quinolinato)iridium(III) (abbreviation: [Ir(bzq).sub.3]), tris(2-phenylquinolinato-N,C.sup.2')iridium(III) (abbreviation: [Ir(pq).sub.3]), bis(2-phenylquinolinato-N,C.sup.2')iridium(III) acetylacetonate (abbreviation: [Ir(pq).sub.2(acac)]), bis[2-(2-pyridinyl-κN)phenyl-κC][2-(4-phenyl-2-pyridinyl-κN)phenyl-κC]iridium(III) (abbreviation: [Ir(ppy).sub.2(4dppy)]), and bis[2-(2-pyridinyl-κN)phenyl-κC][2-(4-methyl-5-phenyl-2-pyridinyl-κN)phenyl-κC]; organometallic complexes such as bis(2,4-diphenyl-1,3-oxazolato-N,C.sup.2')iridium(III) acetylacetonate (abbreviation: [Ir(dpo).sub.2(acac)]), bis{2-[4'-(perfluorophenyl)phenyl]pyridinato-N,C.sup.2'}iridium(III) acetylacetonate (abbreviation: [Ir(p-PF-ph).sub.2(acac)]), and bis(2-phenylbenzothiazolato-N,C.sup.2')iridium(III) acetylacetonate (abbreviation: [Ir(bt).sub.2(acac)]); and rare earth metal complexes such as tris(acetylacetonato)(monophenanthroline)terbium(III) (abbreviation: [Tb(acac).sub.3(Phen)]) can be given.

(62) As a phosphorescent material that exhibits yellow or red and whose emission spectrum has a peak wavelength at greater than or equal to 570 nm and less than or equal to 750 nm, the following substances can be given.

(63) For example, organometallic complexes having a pyrimidine skeleton, such as (diisobutyrylmethanato)bis[4,6-bis(3-methylphenyl)pyrimidinato]iridium(III) (abbreviation: [Ir(5mdppm).sub.2(dibm)]), bis[4,6-bis(3-methylphenyl)pyrimidinato](dipivaloylmethanato)iridium(III) (abbreviation: [Ir(5mdppm).sub.2(dpm)]), and (dipivaloylmethanato)bis[4,6-di(naphthalen-1-yl)pyrimidinato]iridium(III) (abbreviation: [Ir(dlnpm).sub.2(dpm)]); organometallic complexes having a pyrazine skeleton, such as (acetylacetonato)bis(2,3,5-triphenylpyrazinato)iridium(III) (abbreviation: [Ir(tppr).sub.2(acac)]), bis(2,3,5-triphenylpyrazinato)(dipivaloylmethanato)iridium(III) (abbreviation: [Ir(tppr).sub.2(dpm)]), bis{4,6-dimethyl-2-[3-(3,5-dimethylphenyl)-5-phenyl-2-pyrazinyl-N]phenyl-κC}(2,6-dimethyl-3,5-heptanedionato-κ.sup.2O,O')iridium(III) (abbreviation: [Ir(dmdppr-P).sub.2(dibm)]), bis{4,6-dimethyl-2-[5-(4-cyano-2,6-dimethylphenyl)-3-(3,5-dimethylphenyl)-2-pyrazinyl-κN]phenyl-κC}(2,2,6,6-tetramethyl-3,5-heptanedionato-κ.sup.2O,O')iridium(III) (abbreviation: [Ir(dmdppr-dmCP).sub.2(dpm)]), (acetylacetonato)bis[2-methyl-3-phenylquinoxalinato-N,C.sup.2']iridium(III) (abbreviation: [Ir(mpq).sub.2(acac)]), (acetylacetonato)bis(2,3-diphenylquinoxalinato-N,C.sup.2')iridium(III) (abbreviation: [Ir(dpq).sub.2(acac)]), and (acetylacetonato)bis[2,3-bis(4-fluorophenyl)quinoxalinato]iridium(III) (abbreviation: [Ir(Fdpq).sub.2(acac)]); organometallic complexes having a pyridine skeleton, such as tris(1-phenylisoquinolinato-N,C.sup.2')iridium(III) (abbreviation: [Ir(piq).sub.3]), bis(1-phenylisoquinolinato-N,C.sup.2')iridium(III) acetylacetonate (abbreviation: [Ir(piq).sub.2(acac)]), and bis[4,6-dimethyl-2-(2-quinolinyl-κN)phenyl-κC](2,4-pentanedionato-κ.sup.2O,O')iridium(III) (abbreviation: [Ir(dmpqn).sub.2(acac)]); platinum complexes such as 2,3,7,8,12,13,17,18-octaethyl-21H,23H-porphyrinplatinum(II) (abbreviation: [PtOEP]); and rare earth metal complexes such as

tris(1,3-diphenyl-1,3-propanedionato)(monophenanthroline)europium(III) (abbreviation: [Eu(DBM).sub.3(Phen)]) and tris[1-(2-thenoyl)-3,3,3-trifluoroacetato] (monophenanthroline)europium(III) (abbreviation: [Eu(TTA).sub.3(Phen)]) can be given.

(64) Shown below are specific examples of the organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton, which is an electron-transport material shown above as a material that can be used for the light-emitting layer **113**. Note that any of these electron-transport materials can also be used for the electron-transport layer **114** described later.

(65) ##STR00009## ##STR00010## ##STR00011## ##STR00012## ##STR00013##
##STR00014## ##STR00015## ##STR00016##

(66) In addition to the above organic compounds, other examples of the organic compound that can be used for the light-emitting layer **113** are given below (some of them overlap with the above) in terms of favorable combination with a light-emitting substance (a fluorescent material or a phosphorescent material).

(67) In the case where the light-emitting substance is a fluorescent material, examples of the organic compound that is favorably combined with the fluorescent material include condensed polycyclic aromatic compounds, such as an anthracene derivative, a tetracene derivative, a phenanthrene derivative, a pyrene derivative, a chrysene derivative, and a dibenzo[g,p]chrysene derivative.

(68) Specific examples of the organic compound (the host material) that is favorably combined with the fluorescent material include 9-phenyl-3-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazole (abbreviation: PCzPA), 3,6-diphenyl-9-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazole (abbreviation: DPCzPA), 3-[4-(1-naphthyl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPN), 9,10-diphenylanthracene (abbreviation: DPAnth), N,N-diphenyl-9-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazol-3-amine (abbreviation: CzA1PA), 4-(10-phenyl-9-anthryl)triphenylamine (abbreviation: DPhPA), YGAPA, PCAPA, N,9-diphenyl-N-{4-[4-(10-phenyl-9-anthryl)phenyl]phenyl}-9H-carbazol-3-amine (abbreviation: PCAPBA), N-(9,10-diphenyl-2-anthryl)-N,9-diphenyl-9H-carbazol-3-amine (abbreviation: 2PCAPA), 6,12-dimethoxy-5,11-diphenylchrysene, N,N,N,N,N',N',N'',N''-octaphenyldibenzo[g,p]chrysene-2,7,10,15-tetraamine (abbreviation: DBC1), 9-[4-(10-phenyl-9-anthracenyl)phenyl]-9H-carbazole (abbreviation: CzPA), 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA), 6-[3-(9,10-diphenyl-2-anthryl)phenyl]-benzo[b]naphtho[1,2-d]furan (abbreviation: 2mBnfPPA), 9-phenyl-10-{4-(9-phenyl-9H-fluoren-9-yl)-biphenyl-4'-yl}-anthracene (abbreviation: FLPPA), 9,10-bis(3,5-diphenylphenyl)anthracene (abbreviation: DPPA), 9,10-di(2-naphthyl)anthracene (abbreviation: DNA), 2-tert-butyl-9,10-di(2-naphthyl)anthracene (abbreviation: t-BuDNA), 9,9'-bianthryl (abbreviation: BANT), 9,9'-(stilbene-3,3'-diyl)diphenanthrene (abbreviation: DPNS), 9,9'-(stilbene-4,4'-diyl)diphenanthrene (abbreviation: DPNS2), 1,3,5-tri(1-pyrenyl)benzene (abbreviation: TPB3), 5,12-diphenyltetracene, and 5,12-bis(biphenyl-2-yl)tetracene.

(69) In the case where the light-emitting substance is a phosphorescent material, an organic compound having triplet excitation energy (energy difference between a ground state and a triplet excited state) higher than that of the light-emitting substance is preferably selected as the organic compound that is favorably combined with the phosphorescent material. Note that in the case where a plurality of organic compounds (e.g., a first host material and a second host material (or an assist material)) are used in combination with a light-emitting substance in order to form an exciplex, the plurality of organic compounds are preferably mixed with a phosphorescent material.

(70) Such a structure makes it possible to efficiently obtain light emission utilizing ExTET (Exciplex-Triplet Energy Transfer), which is energy transfer from an exciplex to a light-emitting substance. Note that a combination of the plurality of organic compounds that easily forms an exciplex is preferably employed, and it is particularly preferable to combine a compound that easily

accepts holes (hole-transport material) and a compound that easily accepts electrons (electron-transport material).

(71) In the case where the light-emitting substance is a phosphorescent material, examples of the organic compounds (the host material and the assist material) that is favorably combined with the phosphorescent material include an aromatic amine, a carbazole derivative, a dibenzothiophene derivative, a dibenzofuran derivative, a zinc- or aluminum-based metal complex, an oxadiazole derivative, a triazole derivative, a benzimidazole derivative, a quinoxaline derivative, a dibenzoquinoxaline derivative, a pyrimidine derivative, a pyrazine derivative, a triazine derivative, a pyridine derivative, a bipyridine derivative, and a phenanthroline derivative.

(72) Specific examples include triazole derivatives such as 2-(4-biphenyl)-5-(4-tert-butylphenyl)-1,3,4-oxadiazole (abbreviation: PBD), 1,3-bis[5-(p-tert-butylphenyl)-1,3,4-oxadiazol-2-yl]benzene (abbreviation: OXD-7), 9-[4-(5-phenyl-1,3,4-oxadiazol-2-yl)phenyl]-9H-carbazole (abbreviation: CO11), 3-(4-biphenyl)-4-phenyl-5-(4-tert-butylphenyl)-1,2,4-triazole (abbreviation: TAZ), and 3-(4-tert-butylphenyl)-4-(4-ethylphenyl)-5-(4-biphenyl)-1,2,4-triazole (abbreviation: p-EtTAZ); and quinoxaline derivatives or dibenzoquinoxaline derivatives, such as 2,2',2''-(1,3,5-benzenetriyl)tris(1-phenyl-1H-benzimidazole) (abbreviation: TPBI), 2-[3-(dibenzothiophen-4-yl)phenyl]-1-phenyl-1H-benzimidazole (abbreviation: mDBTBIIm-II), 4,4'-bis(5-methylbenzoxazol-2-yl)stilbene (abbreviation: BzOs), bathophenanthroline (abbreviation: Bphen), bathocuproine (abbreviation: BCP), 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBphen), 2-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 2mDBTPDBq-II), 2-[3'-(dibenzothiophen-4-yl)biphenyl-3-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mDBTBPDq-II), 2-[3'-(9H-carbazol-9-yl)biphenyl-3-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mCzBPDBq), 2-[4-(3,6-diphenyl-9H-carbazol-9-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 2CzPDBq-III), 7-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 7mDBTPDBq-II), and 6-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 6mDBTPDBq-II).

(73) Specific examples further include pyrimidine derivatives such as 4,6-bis[3-(phenanthren-9-yl)phenyl]pyrimidine (abbreviation: 4,6mPnP2Pm), 4,6-bis[3-(4-dibenzothienyl)phenyl]pyrimidine (abbreviation: 4,6mDBTP2Pm-II), 4,6-bis[3-(9H-carbazol-9-yl)phenyl]pyrimidine (abbreviation: 4,6mCzP2Pm); triazine derivatives such as 2-{4-[3-(N-phenyl-9H-carbazol-3-yl)-9H-carbazol-9-yl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: PCCzPTzn); triazine derivatives such as 9-[3-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl]-9'-phenyl-2,3'-bi-9H-carbazole (abbreviation: mPCCzPTzn-02); and pyridine derivatives such as 3,5-bis[3-(9H-carbazol-9-yl)phenyl]pyridine (abbreviation: 35DCzPPy), and 1,3,5-tri[3-(3-pyridyl)phenyl]benzene (abbreviation: TmPyPB).

(74) Furthermore, a high-molecular compound such as poly(2,5-pyridinediyl) (abbreviation: PPy), poly[(9,9-dihexylfluorene-2,7-diyl)-co-(pyridine-3,5-diyl)] (abbreviation: PF-Py), or poly[(9,9-dioctylfluorene-2,7-diyl)-co-(2,2'-bipyridine-6,6'-diyl)] (abbreviation: PF-BPy) can be used.

(75) In the case where a plurality of organic compounds are used in the light-emitting layer **113**, two kinds of compounds that form an exciplex (a first organic compound and a second organic compound) and a light-emitting substance may be mixed and used. In this case, various organic compounds can be used in appropriate combination; to form an exciplex efficiently, it is particularly preferable to combine a compound that easily accepts holes (hole-transport material) and a compound that easily accepts electrons (electron-transport material). Note that, as specific examples of the hole-transport material and the electron-transport material, the materials described in this embodiment can be used. With the structure, high efficiency, low voltage, and a long lifetime can be achieved at the same time.

(76) The TADF material is a material that can up-convert a triplet excited state into a singlet excited state (reverse intersystem crossing) using a little thermal energy and efficiently exhibits light emission (fluorescence) from the singlet excited state. The thermally activated delayed fluorescence is efficiently obtained under the condition where the difference in energy between the

triplet excited level and the singlet excited level is greater than or equal to 0 eV and less than or equal to 0.2 eV, preferably greater than or equal to 0 eV and less than or equal to 0.1 eV. Note that delayed fluorescence by the TADF material refers to light emission having a spectrum similar to that of normal fluorescence and an extremely long lifetime. The lifetime is 10^{-6} seconds or longer, preferably 10^{-3} seconds or longer.

(77) Examples of the TADF material include fullerene, a derivative thereof, an acridine derivative such as proflavine, and eosin. Other examples include a metal-containing porphyrin such as a porphyrin containing magnesium (Mg), zinc (Zn), cadmium (Cd), tin (Sn), platinum (Pt), indium (In), or palladium (Pd). Examples of the metal-containing porphyrin include a protoporphyrin-tin fluoride complex (abbreviation: SnF.sub.2(Proto IX)), a mesoporphyrin-tin fluoride complex (abbreviation: SnF.sub.2(Meso IX)), a hematoporphyrin-tin fluoride complex (abbreviation: SnF.sub.2(Hemato IX)), a coproporphyrin tetramethyl ester-tin fluoride complex (abbreviation: SnF.sub.2(Copro III-4Me)), an octaethylporphyrin-tin fluoride complex (abbreviation: SnF.sub.2(OEP)), an etioporphyrin-tin fluoride complex (abbreviation: SnF.sub.2(Etio I)), and an octaethylporphyrin-platinum chloride complex (abbreviation: PtCl.sub.2OEP).

(78) Alternatively, a heterocyclic compound having a π -electron rich heteroaromatic ring and a π -electron deficient heteroaromatic ring, such as 2-(biphenyl-4-yl)-4,6-bis(12-phenylindolo[2,3-a]carbazol-11-yl)-1,3,5-triazine (abbreviation: PIC-TRZ), 2-{4-[3-(N-phenyl-9H-carbazol-3-yl)-9H-carbazol-9-yl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: PCCzPTzn), 2-[4-(10H-phenoxazin-10-yl)phenyl]-4,6-diphenyl-1,3,5-triazine (abbreviation: PXZ-TRZ), 3-[4-(5-phenyl-5,10-dihydrophenazin-10-yl)phenyl]-4,5-diphenyl-1,2,4-triazole (abbreviation: PPZ-3TPT), 3-(9,9-dimethyl-9H-acridin-10-yl)-9H-xanthen-9-one (abbreviation: ACRXTN), bis[4-(9,9-dimethyl-9,10-dihydroacridine)phenyl]sulfone (abbreviation: DMAC-DPS), or 10-phenyl-10H,10'H-spiro[acridin-9,9'-anthracen]-10'-one (abbreviation: ACRSA) may be used.

(79) Note that a substance in which a π -electron rich heteroaromatic ring is directly bonded to a π -electron deficient heteroaromatic ring is particularly preferable because both the donor property of the π -electron rich heteroaromatic ring and the acceptor property of the π -electron deficient heteroaromatic ring are improved and the energy difference between the singlet excited state and the triplet excited state becomes small.

(80) Note that the TADF material can also be used in combination with another organic compound. In particular, the TADF material can be combined with the host materials, the hole-transport materials, and the electron-transport materials described above, and the organic compound of one embodiment of the present invention described in Embodiment 1 is preferably used as a host material for the TADF material.

(81) Any of the above materials may be used in combination with a low-molecular material or a high-molecular material. For the deposition, a known method (a vacuum evaporation method, a coating method, a printing method, or the like) can be used as appropriate.

(82) <Electron-Transport Layer>

(83) The electron-transport layer **114** is a layer transporting electrons, which are injected from the second electrode **102** through the electron-injection layer **115** to be described later, to the light-emitting layer **113**. Note that the electron-transport layer **114** is a layer containing an electron-transport material. As the electron-transport material, a substance having an electron mobility of greater than or equal to 1×10^{-6} cm²/Vs is preferable. Note that other substances can be used as long as they have a property of transporting more electrons than holes. The electron-transport layers (**114**, **114a**, and **114b**) each function even with a single-layer structure, but can improve the device characteristics when having a stacked-layer structure of two or more layers as needed.

(84) As the organic compound that can be used for the electron-transport layer **114**, it is possible to use, in addition to the electron-transport materials (the organic compounds having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton) specifically given above

as examples that can be used in the light-emitting layer **113**, a material having a high electron-transport property (an electron-transport material), such as a metal complex having a quinoline skeleton, a metal complex having a benzoquinoline skeleton, a metal complex having an oxazole skeleton, a metal complex having a thiazole skeleton, an oxadiazole derivative, a triazole derivative, an imidazole derivative, an oxazole derivative, a thiazole derivative, a phenanthroline derivative, a quinoline derivative having a quinoline ligand, a benzoquinoline derivative, a quinoxaline derivative, a dibenzoquinoxaline derivative, a pyridine derivative, a bipyridine derivative, a pyrimidine derivative, or a π -electron deficient heteroaromatic compound such as a nitrogen-containing heteroaromatic compound.

(85) Specific examples of the electron-transport material (other than the organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton) include metal complexes having a quinoline skeleton or a benzoquinoline skeleton, such as tris(8-quinolinolato)aluminum(III) (abbreviation: Alq.sub.3), tris(4-methyl-8-quinolinolato)aluminum(III) (abbreviation: Almq.sub.3), bis(10-hydroxybenzo[h]quinolinato)beryllium(II) (abbreviation: BeBq.sub.2), bis(2-methyl-8-quinolinolato)(4-phenylphenolato)aluminum(III) (abbreviation: BAlq), and bis(8-quinolinolato)zinc(II) (abbreviation: Znq), and metal complexes having an oxazole skeleton or a thiazole skeleton, such as bis[2-(2-benzoxazolyl)phenolato]zinc(II) (abbreviation: ZnPBO), bis[2-(2-benzothiazolyl)phenolato]zinc(II) (abbreviation: ZnBTZ), and bis[2-(2-hydroxyphenyl)benzothiazolato]zinc(II) (abbreviation: Zn(BTZ).sub.2).

(86) Other than the metal complexes, it is possible to use oxadiazole derivatives such as PBD, OXD-7, and CO11, triazole derivatives such as TAZ and p-EtTAZ, imidazole derivatives (including benzimidazole derivatives) such as TPBI and mDBTBIIm-II, an oxazole derivative such as BzOs, phenanthroline derivatives such as Bphen, BCP, and NBphen, quinoxaline derivatives and dibenzoquinoxaline derivatives, such as 2mDBTPDBq-II, 2mDBTBPDBq-II, 2mCzBPDBq, 2CzPDBq-III, 7mDBTPDBq-II, and 6mDBTPDBq-II, pyridine derivatives such as 35DCzPPy and TmPyPB, pyrimidine derivatives such as 4,6mPnP2Pm, 4,6mDBTP2Pm-II, and 4,6mCzP2Pm, and triazine derivatives such as PCCzPTzn and mPCCzPTzn-02.

(87) It is also possible to use high-molecular compounds such as PPy, PF-Py, and PF-BPy.

(88) Note that for the electron-transport layer **114** of the light-emitting device of one embodiment of the present invention, it is preferable to use an electron-transport material (an organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton) in combination with an alkali metal or an alkaline earth metal, a compound thereof, or the like. Such a combination enables the electron-transport layer **114** to function also as an electron-injection layer; thus, a light-emitting device in which the electron-injection layer **115** is not formed between the electron-transport layer **114** and the second electrode **102** can have substantially the same characteristics and reliability as in the case where the electron-injection layer **115** is formed.

(89) Specific examples of the alkali metal or the alkaline earth metal include lithium (Li), cesium (Cs), magnesium (Mg), and calcium (Ca). As the compound of an alkali metal or an alkaline earth metal, it is possible to use an inorganic compound such as lithium oxide or cesium carbonate or an organic compound such as 8-(quinolinolato)lithium (abbreviation: Liq), 2-(2-pyridyl)phenolatolithium (abbreviation: LiPP), 2-(2-pyridyl)-3-pyridinolato lithium (abbreviation: LiPPy), or 4-phenyl-2-(2-pyridyl)phenolatolithium (abbreviation: LiPPP).

(90) In the light-emitting device of another embodiment of the present invention, which uses for the light-emitting layer **113** an organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton, sufficient characteristics can be obtained even when the electron-transport layer **114** has a single-layer structure.

(91) <Electron-Injection Layer>

(92) The electron-injection layer **115** is a layer for increasing the efficiency of electron injection from the cathode; thus, the electron-injection layer **115** is preferably formed using a material whose

LUMO level value has a small difference (0.5 eV or less) from the work function value of a cathode material. Thus, the electron-injection layer **115** can be formed using an alkali metal, an alkaline earth metal, or a compound thereof, such as lithium, cesium, lithium fluoride (LiF), cesium fluoride (CsF), calcium fluoride (CaF₂), 8-(quinolinolato)-lithium (abbreviation: Liq), 2-(2-pyridyl)phenolatolithium (abbreviation: LiPP), 2-(2-pyridyl)-3-pyridinolitolithium (abbreviation: LiPPy), 4-phenyl-2-(2-pyridyl)phenolatolithium (abbreviation: LiPPP), lithium oxide (LiO_x), or cesium carbonate. A rare earth metal compound like erbium fluoride (ErF₃) can also be used.

(93) When a charge-generation layer **104** is provided between two EL layers (**103a** and **103b**) as in the light-emitting device illustrated in FIG. **1B**, a structure in which a plurality of EL layers are stacked between the pair of electrodes (also referred to as a tandem structure) can be employed. Note that in this embodiment, functions and materials of the hole-injection layer (**111**), the hole-transport layer (**112**), the light-emitting layer (**113**), the electron-transport layer (**114**), and the electron-injection layer (**115**) that are illustrated in FIG. **1A** are the same as those of hole-injection layer (**111a** and **111b**), hole-transport layers (**112a** and **112b**), light-emitting layers (**113a** and **113b**), electron-transport layers (**114a** and **114b**), and electron-injection layers (**115a** and **115b**) that are illustrated in FIG. **1B**.

(94) <Charge-Generation Layer>

(95) In the light-emitting device of FIG. **1B**, the charge-generation layer **104** has a function of injecting electrons into the EL layer **103a** and injecting holes into the EL layer **103b** when voltage is applied between the first electrode (anode) **101** and the second electrode (cathode) **102**. Note that the charge-generation layer **104** may have either a structure in which an electron acceptor (acceptor) is added to a hole-transport material or a structure in which an electron donor (donor) is added to an electron-transport material. Alternatively, both of these structures may be stacked. Note that forming the charge-generation layer **104** with the use of any of the above materials can suppress an increase in drive voltage in the case where the EL layers are stacked.

(96) In the case where the charge-generation layer **104** has a structure in which an electron acceptor is added to a hole-transport material, any of the materials described in this embodiment can be used as the hole-transport material. As the electron acceptor, 7,7,8,8-tetracyano-2,3,5,6-tetrafluoroquinodimethane (abbreviation: F₄-TCNQ), chloranil, and the like can be given. Other examples include oxides of metals belonging to Group 4 to Group 8 of the periodic table. Specific examples are vanadium oxide, niobium oxide, tantalum oxide, chromium oxide, molybdenum oxide, tungsten oxide, manganese oxide, and rhenium oxide.

(97) In the case where the charge-generation layer **104** has a structure in which an electron donor is added to an electron-transport material, any of the materials described in this embodiment can be used as the electron-transport material. As the electron donor, it is possible to use an alkali metal, an alkaline earth metal, a rare earth metal, metals belonging to Groups 2 and 13 of the periodic table, or an oxide or carbonate thereof. Specifically, lithium (Li), cesium (Cs), magnesium (Mg), calcium (Ca), ytterbium (Yb), indium (In), lithium oxide, cesium carbonate, or the like is preferably used. An organic compound such as tetrathianaphthacene may be used as the electron donor.

(98) Although FIG. **1B** illustrates the structure in which two EL layers **103** are stacked, a structure may be employed in which three or more EL layers are stacked with a charge generation layer provided between different EL layers.

(99) <Substrate>

(100) The light-emitting device described in this embodiment can be formed over any of a variety of substrates. Note that the type of the substrate is not limited to a certain type. Examples of the substrate include semiconductor substrates (e.g., a single crystal substrate and a silicon substrate), an SOI substrate, a glass substrate, a quartz substrate, a plastic substrate, a metal substrate, a stainless steel substrate, a substrate including stainless steel foil, a tungsten substrate, a substrate

including tungsten foil, a flexible substrate, a laminate film, paper including a fibrous material, and a base material film.

(101) Note that examples of the glass substrate include barium borosilicate glass, aluminoborosilicate glass, and soda lime glass. Examples of the flexible substrate, the laminate film, and the base material film include plastics typified by polyethylene terephthalate (PET), polyethylene naphthalate (PEN), and polyether sulfone (PES); a Synthesis resin such as an acrylic resin; polypropylene; polyester; polyvinyl fluoride; polyvinyl chloride; polyamide; polyimide; an aramid resin; an epoxy resin; an inorganic vapor deposition film; and paper.

(102) For fabrication of the light-emitting device in this embodiment, a vacuum process such as an evaporation method or a solution process such as a spin coating method or an ink-jet method can be used. In the case of using an evaporation method, a physical vapor deposition method (PVD method) such as a sputtering method, an ion plating method, an ion beam evaporation method, a molecular beam evaporation method, or a vacuum evaporation method, a chemical vapor deposition method (CVD method), or the like can be used. Specifically, the functional layers (the hole-injection layers (**111**, **111a**, and **111b**), the hole-transport layers (**112**, **112a**, and **112b**), the light-emitting layers (**113**, **113a**, **113b**, and **113c**), the electron-transport layers (**114**, **114a**, and **114b**), and the electron-injection layers (**115**, **115a**, and **115b**)) included in the EL layers and the charge-generation layers (**104**, **104a**, and **104b**) of the light-emitting device can be formed by an evaporation method (e.g., a vacuum evaporation method), a coating method (e.g., a dip coating method, a die coating method, a bar coating method, a spin coating method, or a spray coating method), a printing method (e.g., an ink-jet method, a screen printing (stencil) method, an offset printing (planography) method, a flexography (relief printing) method, a gravure printing method, a micro-contact printing method, or a nanoinprinting method), or the like.

(103) Note that materials that can be used for the functional layers (the hole-injection layers (**111**, **111a**, and **111b**), the hole-transport layers (**112**, **112a**, and **112b**), the light-emitting layers (**113**, **113a**, **113b**, and **113c**), the electron-transport layers (**114**, **114a**, and **114b**), and the electron-injection layers (**115**, **115a**, and **115b**)) included in the EL layers (**103**, **103a**, and **103b**) and the charge-generation layers (**104**, **104a**, and **104b**) of the light-emitting device described in this embodiment are not limited to the above materials, and other materials can also be used in combination as long as the functions of the layers are fulfilled. For example, a high molecular compound (e.g., an oligomer, a dendrimer, and a polymer), a middle molecular compound (a compound between a low molecular compound and a high molecular compound with a molecular weight of 400 to 4000), or an inorganic compound (e.g., a quantum dot material) can be used. As the quantum dot material, a colloidal quantum dot material, an alloyed quantum dot material, a core-shell quantum dot material, a core quantum dot material, or the like can be used.

(104) The structure described in this embodiment can be used in an appropriate combination with any of the structures described in the other embodiments.

Embodiment 2

(105) In this embodiment, a light-emitting apparatus of one embodiment of the present invention will be described. Note that a light-emitting apparatus illustrated in FIG. 2A is an active-matrix light-emitting apparatus in which transistors (FETs) **202** over a first substrate **201** are electrically connected to light-emitting devices (**203R**, **203G**, **203B**, and **203W**); the light-emitting devices (**203R**, **203G**, **203B**, and **203W**) include a common EL layer **204** and each have a microcavity structure in which the optical path length between electrodes of each light-emitting device is adjusted according to the emission color of the light-emitting device. In addition, the light-emitting apparatus is a top-emission light-emitting apparatus in which light is emitted from the EL layer **204** through color filters (**206R**, **206G**, and **206B**) formed on a second substrate **205**.

(106) In the light-emitting apparatus illustrated in FIG. 2A, a first electrode **207** is formed so as to function as a reflective electrode. A second electrode **208** is formed so as to function as a semi-transmissive and semi-reflective electrode. Note that description in any of the other embodiments

can be referred to for electrode materials forming the first electrode **207** and the second electrode **208** and appropriate materials can be used.

(107) In the case where the light-emitting device **203R** is a red-light-emitting device, the light-emitting device **203G** is a green-light-emitting device, the light-emitting device **203B** is a blue-light-emitting device, and the light-emitting device **203W** is a white-light-emitting device in FIG. 2A, for example, a gap between the first electrode **207** and the second electrode **208** in the light-emitting device **203R** is adjusted to have an optical path length **200R**, a gap between the first electrode **207** and the second electrode **208** in the light-emitting device **203G** is adjusted to have an optical path length **200G**, and a gap between the first electrode **207** and the second electrode **208** in the light-emitting device **203B** is adjusted to have an optical path length **200B** as illustrated in FIG. 2B. Note that optical adjustment can be performed in such a manner that a conductive layer **210R** is stacked over the first electrode **207** in the light-emitting device **203R** and a conductive layer **210G** is stacked over the first electrode **207** in the light-emitting device **203G** as illustrated in FIG. 2B.

(108) The color filters (**206R**, **206G**, and **206B**) are formed on the second substrate **205**. Note that the color filters are filters each transmitting visible light in a specific wavelength range and blocking visible light in a specific wavelength range. Thus, as illustrated in FIG. 2A, the color filter **206R** that transmits only light in the red wavelength range is provided in a position overlapping with the light-emitting device **203R**, whereby red light emission can be obtained from the light-emitting device **203R**. The color filter **206G** that transmits only light in the green wavelength range is provided in a position overlapping with the light-emitting device **203G**, whereby green light emission can be obtained from the light-emitting device **203G**. The color filter **206B** that transmits only light in the blue wavelength range is provided in a position overlapping with the light-emitting device **203B**, whereby blue light emission can be obtained from the light-emitting device **203B**. Note that the light-emitting device **203W** can emit white light without a color filter. Note that a black layer (black matrix) **209** may be provided at an end portion of one type of color filter. The color filters (**206R**, **206G**, and **206B**) and the black layer **209** may be covered with an overcoat layer using a transparent material.

(109) Although the light-emitting device illustrated in FIG. 2A has a structure in which light is extracted from the second substrate **205** side (top emission structure), the light-emitting device may have a structure in which light is extracted from the first substrate **201** side where the FETs **202** are formed (bottom emission structure) as illustrated in FIG. 2C. In the case of a bottom-emission light-emitting device, the first electrode **207** is formed so as to function as a semi-transmissive and semi-reflective electrode and the second electrode **208** is formed so as to function as a reflective electrode. As the first substrate **201**, a substrate having at least a light-transmitting property is used. As illustrated in FIG. 2C, color filters (**206R'**, **206G'**, and **206B'**) are provided closer to the first substrate **201** than the light-emitting devices (**203R**, **203G**, and **203B**) are.

(110) FIG. 2A illustrates the case where the light-emitting devices are the red-light-emitting device, the green-light-emitting device, the blue-light-emitting device, and the white-light-emitting device; however, the light-emitting devices of embodiments of the present invention are not limited to the above structures, and a yellow-light-emitting device or an orange-light-emitting device may be included. Note that description in any of the other embodiments can be referred to for materials that are used for the EL layers (a light-emitting layer, a hole-injection layer, a hole-transport layer, an electron-transport layer, an electron-injection layer, a charge-generation layer, and the like) to fabricate each of the light-emitting devices and appropriate materials can be used. In that case, a color filter needs to be appropriately selected according to the emission color of the light-emitting device.

(111) With the above structure, a light-emitting apparatus including light-emitting devices that exhibit a plurality of emission colors can be obtained.

(112) Note that the structures described in this embodiment can be used in an appropriate combination with any of the structures described in the other embodiments.

Embodiment 3

(113) In this embodiment, a light-emitting apparatus of one embodiment of the present invention will be described.

(114) The use of the device structure of the light-emitting device of one embodiment of the present invention allows fabrication of an active-matrix light-emitting apparatus or a passive-matrix light-emitting apparatus. Note that an active-matrix light-emitting apparatus has a structure including a combination of a light-emitting device and a transistor (FET). Thus, each of a passive-matrix light-emitting apparatus and an active-matrix light-emitting apparatus is included in one embodiment of the present invention. Note that any of the light-emitting devices described in the other embodiments can be used in the light-emitting apparatus described in this embodiment.

(115) In this embodiment, an active-matrix light-emitting apparatus will be described with reference to FIG. 3.

(116) FIG. 3A is a top view illustrating a light-emitting apparatus **21**, and FIG. 3B is a cross-sectional view taken along a chain line A-A' in FIG. 3A. The active-matrix light-emitting apparatus includes a pixel portion **302**, a driver circuit portion (source line driver circuit) **303**, and driver circuit portions (gate line driver circuits) (**304a** and **304b**) that are provided over a first substrate **301**. The pixel portion **302** and the driver circuit portions (**303**, **304a**, and **304b**) are sealed between the first substrate **301** and a second substrate **306** with a sealant **305**.

(117) A lead wiring **307** is provided over the first substrate **301**. The lead wiring **307** is electrically connected to an FPC **308** that is an external input terminal. Note that the FPC **308** transmits a signal (e.g., a video signal, a clock signal, a start signal, or a reset signal) or a potential from the outside to the driver circuit portions (**303**, **304a**, and **304b**). The FPC **308** may be provided with a printed wiring board (PWB). Note that the light-emitting apparatus provided with an FPC or a PWB is included in the category of a light-emitting apparatus.

(118) Next, FIG. 3B illustrates a cross-sectional structure of the light-emitting apparatus.

(119) The pixel portion **302** is made up of a plurality of pixels each of which includes an FET (switching FET) **311**, an FET (current control FET) **312**, and a first electrode **313** electrically connected to the FET **312**. Note that the number of FETs included in each pixel is not particularly limited and can be set appropriately as needed.

(120) As FETs **309**, **310**, **311**, and **312**, for example, a staggered transistor or an inverted staggered transistor can be used without particular limitation. A top-gate transistor, a bottom-gate transistor, or the like may be used.

(121) Note that there is no particular limitation on the crystallinity of a semiconductor that can be used for the FETs **309**, **310**, **311**, and **312**, and an amorphous semiconductor or a semiconductor having crystallinity (a microcrystalline semiconductor, a polycrystalline semiconductor, a single crystal semiconductor, or a semiconductor partly including crystal regions) may be used. The use of a semiconductor having crystallinity can suppress deterioration of the transistor characteristics, which is preferable.

(122) For these semiconductors, a Group 14 element, a compound semiconductor, an oxide semiconductor, an organic semiconductor, or the like can be used, for example. Typically, a semiconductor containing silicon, a semiconductor containing gallium arsenide, an oxide semiconductor containing indium, or the like can be used.

(123) The driver circuit portion **303** includes the FET **309** and the FET **310**. The FET **309** and the FET **310** may be formed with a circuit including transistors having the same conductivity type (either n-channel transistors or p-channel transistors) or a CMOS circuit including an n-channel transistor and a p-channel transistor. Furthermore, a structure including a driver circuit outside may be employed.

(124) An end portion of the first electrode **313** is covered with an insulator **314**. For the insulator **314**, an organic compound such as a negative photosensitive resin or a positive photosensitive resin (acrylic resin), or an inorganic compound such as silicon oxide, silicon oxynitride, or silicon nitride

can be used. An upper end portion or a lower end portion of the insulator **314** preferably has a curved surface with curvature. In that case, favorable coverage with a film formed over the insulator **314** can be obtained.

(125) An EL layer **315** and a second electrode **316** are stacked over the first electrode **313**. The EL layer **315** includes a light-emitting layer, a hole-injection layer, a hole-transport layer, an electron-transport layer, an electron-injection layer, a charge-generation layer, and the like.

(126) The structure and materials described in any of the other embodiments can be used for the structure of a light-emitting device **317** described in this embodiment. Although not illustrated here, the second electrode **316** is electrically connected to the FPC **308** that is an external input terminal.

(127) Although the cross-sectional view illustrated in FIG. **3B** illustrates only one light-emitting device **317**, a plurality of light-emitting devices are arranged in a matrix in the pixel portion **302**. Light-emitting devices from which light of three kinds of colors (R, G, and B) are obtained are selectively formed in the pixel portion **302**, whereby a light-emitting apparatus capable of full-color display can be formed. In addition to the light-emitting devices from which light of three kinds of colors (R, G, and B) are obtained, for example, light-emitting devices from which light of white (W), yellow (Y), magenta (M), cyan (C), and the like are obtained may be formed. For example, the light-emitting devices from which light of some of the above colors are obtained are added to the light-emitting devices from which light of three kinds of colors (R, G, and B) are obtained, whereby effects such as an improvement in color purity and a reduction in power consumption can be obtained. Alternatively, a light-emitting apparatus capable of full-color display may be fabricated by a combination with color filters. As the kinds of color filters, red (R), green (G), blue (B), cyan (C), magenta (M), yellow (Y), and the like can be used.

(128) When the second substrate **306** and the first substrate **301** are bonded to each other with the sealant **305**, the FETs (**309**, **310**, **311**, and **312**) and the light-emitting device **317** over the first substrate **301** are provided in a space **318** surrounded by the first substrate **301**, the second substrate **306**, and the sealant **305**. Note that the space **318** may be filled with an inert gas (e.g., nitrogen or argon) or an organic substance (including the sealant **305**).

(129) An epoxy resin or glass frit can be used for the sealant **305**. It is preferable to use a material that is permeable to as little moisture and oxygen as possible for the sealant **305**. As the second substrate **306**, a material that can be used as the first substrate **301** can be similarly used. Thus, any of the various substrates described in the other embodiments can be appropriately used. As the substrate, a glass substrate, a quartz substrate, or a plastic substrate made of FRP (Fiber-Reinforced Plastics), PVF (polyvinyl fluoride), polyester, an acrylic resin, or the like can be used. In the case where glass frit is used for the sealant, the first substrate **301** and the second substrate **306** are preferably glass substrates in terms of adhesion.

(130) In the above manner, the active-matrix light-emitting apparatus can be obtained.

(131) In the case where the active-matrix light-emitting apparatus is formed over a flexible substrate, the FETs and the light-emitting device may be directly formed over the flexible substrate; alternatively, the FETs and the light-emitting device may be formed over a substrate provided with a separation layer and then separated at the separation layer by application of heat, force, laser irradiation, or the like to be transferred to a flexible substrate. For the separation layer, a stack including inorganic films such as a tungsten film and a silicon oxide film, or an organic resin film of polyimide or the like can be used, for example. Examples of the flexible substrate include, in addition to a substrate over which a transistor can be formed, a paper substrate, a cellophane substrate, an aramid film substrate, a polyimide film substrate, a cloth substrate (including a natural fiber (e.g., silk, cotton, or hemp), a synthetic fiber (e.g., nylon, polyurethane, or polyester), a regenerated fiber (e.g., acetate, cupro, rayon, or regenerated polyester), or the like), a leather substrate, and a rubber substrate. With the use of any of these substrates, high durability, high heat resistance, a reduction in weight, and a reduction in thickness can be achieved.

(132) The light-emitting device included in the active-matrix light-emitting apparatus may be

driven with a structure in which pulsed light (with a frequency of kHz or MHz, for example) emitted from the light-emitting device is used for display. The light-emitting device formed using any of the above organic compounds has excellent frequency characteristics; thus, time for driving the light-emitting device can be shortened, and thus the power consumption can be reduced. Furthermore, a reduction in driving time leads to inhibition of heat generation, so that the degree of deterioration of the light-emitting device can be reduced.

(133) Note that the structures described in this embodiment can be used in an appropriate combination with the structures described in the other embodiments.

Embodiment 4

(134) In this embodiment, examples of a variety of electronic devices and an automobile completed using the light-emitting device of one embodiment of the present invention or a light-emitting apparatus including the light-emitting device of one embodiment of the present invention are described. Note that the light-emitting apparatus can be used mainly in a display portion of the electronic device described in this embodiment.

(135) Electronic devices illustrated in FIG. 4A to FIG. 4C can include a housing **7000**, a display portion **7001**, a speaker **7003**, an LED lamp **7004**, operation keys **7005** (including a power switch or an operation switch), a connection terminal **7006**, a sensor **7007** (a sensor having a function of measuring force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, chemical substance, sound, time, hardness, electric field, current, voltage, electric power, radiation, flow rate, humidity, gradient, oscillation, odor, or infrared ray), a microphone **7008**, and the like.

(136) FIG. 4A is a mobile computer which can include a switch **7009**, an infrared port **7010**, and the like in addition to the above components.

(137) FIG. 4B is a portable image reproducing device (e.g., a DVD player) which is provided with a recording medium and can include a second display portion **7002**, a recording medium reading portion **7011**, and the like in addition to the above components.

(138) FIG. 4C is a digital camera with a television reception function, which can include an antenna **7014**, a shutter button **7015**, an image receiving portion **7016**, and the like in addition to the above components.

(139) FIG. 4D is a portable information terminal. The portable information terminal has a function of displaying information on three or more surfaces of the display portion **7001**. Here, an example in which information **7052**, information **7053**, and information **7054** are displayed on different surfaces is shown. For example, the user can check the information **7053** displayed in a position that can be observed from above the portable information terminal, with the portable information terminal put in a breast pocket of his/her clothes. The user can see the display without taking out the portable information terminal from the pocket and decide whether to answer the call, for example.

(140) FIG. 4E is a portable information terminal (e.g., a smartphone) and can include the display portion **7001**, the operation key **7005**, and the like in the housing **7000**. Note that a speaker, a connection terminal, a sensor, or the like may be provided in the portable information terminal. The portable information terminal can display characters and image information on its plurality of surfaces. Here, an example is shown in which three icons **7050** are displayed.

(141) Information **7051** indicated by dashed rectangles can be displayed on another surface of the display portion **7001**. Examples of the information **7051** include notification of reception of an e-mail, SNS, or an incoming call, the title and sender of an e-mail, SNS, or the like, the date, the time, remaining battery, and the reception strength of an antenna. Alternatively, the icon **7050** or the like may be displayed in the position where the information **7051** is displayed.

(142) FIG. 4F is a large-size television set (also referred to as TV or a television receiver), which can include the housing **7000**, the display portion **7001**, and the like. In addition, shown here is a structure where the housing **7000** is supported by a stand **7018**. The television set can be operated

with a separate remote controller **7111** or the like. Note that the display portion **7001** may include a touch sensor, in which case the television set may be operated by touch on the display portion **7001** with a finger or the like. The remote controller **7111** may include a display portion for displaying data output from the remote controller **7111**. With operation keys or a touch panel provided in the remote controller **7111**, channels and volume can be operated and images displayed on the display portion **7001** can be operated.

(143) The electronic devices illustrated in FIG. **4A** to FIG. **4F** can have a variety of functions. For example, they can have a function of displaying a variety of data (e.g., a still image, a moving image, and a text image) on the display portion, a touch panel function, a function of displaying a calendar, date, time, or the like, a function of controlling processing with a variety of software (programs), a wireless communication function, a function of being connected to a variety of computer networks with a wireless communication function, a function of transmitting and receiving a variety of data with a wireless communication function, and a function of reading out a program or data stored in a recording medium and displaying it on the display portion. Furthermore, the electronic device including a plurality of display portions can have a function of displaying image data mainly on one display portion while displaying text data mainly on the other display portion, a function of displaying a three-dimensional image by displaying images on a plurality of display portions with a parallax taken into account, or the like. Furthermore, the electronic device including an image receiving portion can have a function of taking a still image, a function of taking a moving image, a function of automatically or manually correcting a taken image, a function of storing a taken image in a recording medium (external or incorporated in the camera), a function of displaying a taken image on the display portion, or the like. Note that functions that the electronic devices illustrated in FIG. **4A** to FIG. **4F** can have are not limited to those, and the electronic devices can have a variety of functions.

(144) FIG. **4G** is a watch-type portable information terminal, which can be used as a smart watch, for example. The watch-type portable information terminal includes the housing **7000**, the display portion **7001**, operation buttons **7022** and **7023**, a connection terminal **7024**, a band **7025**, a microphone **7026**, a sensor **7029**, a speaker **7030**, and the like. The display surface of the display portion **7001** is curved, and display can be performed along the curved display surface. Furthermore, the portable information terminal enables hands-free calling by mutually communicating with, for example, a headset capable of wireless communication. With the connection terminal **7024**, the portable information terminal can perform mutual data transmission with another information terminal and charging. Wireless power feeding can also be employed for the charging operation.

(145) The display portion **7001** mounted in the housing **7000** also serving as a bezel includes a non-rectangular display region. The display portion **7001** can display an icon **7027** indicating time, another icon **7028**, and the like. The display portion **7001** may be a touch panel (input/output device) including a touch sensor (input device).

(146) Note that the smart watch illustrated in FIG. **4G** can have a variety of functions. For example, the smart watch can have a function of displaying a variety of data (e.g., a still image, a moving image, and a text image) on the display portion, a touch panel function, a function of displaying a calendar, date, time, or the like, a function of controlling processing with a variety of software (programs), a wireless communication function, a function of being connected to a variety of computer networks with a wireless communication function, a function of transmitting and receiving a variety of data with a wireless communication function, and a function of reading out a program or data stored in a recording medium and displaying it on the display portion.

(147) Moreover, a speaker, a sensor (a sensor having a function of measuring force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, chemical substance, sound, time, hardness, electric field, current, voltage, electric power, radiation, flow rate, humidity, gradient, oscillation, odor, or infrared rays), a

microphone, and the like can be included inside the housing **7000**.

(148) Note that the light-emitting apparatus of one embodiment of the present invention can be used in the display portions of the electronic devices described in this embodiment, enabling the electronic devices to have a long lifetime.

(149) Another electronic device including the light-emitting apparatus is a foldable portable information terminal illustrated in FIGS. **5A** to **5C**. FIG. **5A** illustrates a portable information terminal **9310** which is opened. FIG. **5B** illustrates the portable information terminal **9310** in a state in the middle of change from one of an opened state and a folded state to the other. FIG. **5C** illustrates the portable information terminal **9310** which is folded. The portable information terminal **9310** is excellent in portability when folded, and is excellent in display browsability when opened because of a seamless large display region.

(150) A display portion **9311** is supported by three housings **9315** joined together by hinges **9313**. Note that the display portion **9311** may be a touch panel (input/output device) including a touch sensor (input device). By bending the display portion **9311** at a portion between two housings **9315** with the use of the hinges **9313**, the portable information terminal **9310** can be reversibly changed in shape from an opened state to a folded state. The light-emitting apparatus of one embodiment of the present invention can be used for the display portion **9311**. In addition, an electronic device having a long lifetime can be achieved. A display region **9312** in the display portion **9311** is a display region that is positioned at a side surface of the portable information terminal **9310** which is folded. On the display region **9312**, information icons, file shortcuts of frequently used applications or programs, and the like can be displayed, and confirmation of information and start of an application can be smoothly performed.

(151) FIGS. **6A** and **6B** illustrate an automobile including the light-emitting apparatus. In other words, the light-emitting apparatus can be integrated into an automobile. Specifically, the light-emitting apparatus can be applied to lights **5101** (including lights of the rear part of the automobile), a wheel **5102**, a part or the whole of a door **5103**, or the like on the outer side of the automobile which is illustrated in FIG. **6A**. The light-emitting apparatus can also be applied to a display portion **5104**, a steering wheel **5105**, a shifter **5106**, a seat **5107**, an inner rearview mirror **5108**, a windshield **5109**, or the like on the inner side of the automobile which is illustrated in FIG. **6B**. The light-emitting apparatus may be used for part of any of the other glass windows.

(152) In the above manner, the electronic devices and automobiles in which the light-emitting apparatus of one embodiment of the present invention is used can be obtained. In that case, an electronic device having a long lifetime can be achieved. In addition, the light-emitting apparatus can be used for electronic devices and automobiles in a variety of fields without being limited to those described in this embodiment.

(153) Note that the structures described in this embodiment can be used in an appropriate combination with any of the structures described in the other embodiments.

Embodiment 5

(154) In this embodiment, the structure of a lighting device fabricated using the light-emitting apparatus of one embodiment of the present invention or the light-emitting device which is part of the light-emitting apparatus will be described with reference to FIG. **7**.

(155) FIGS. **7A** and **7B** each show an example of a cross-sectional view of a lighting device. FIG. **7A** is a bottom-emission lighting device in which light is extracted from the substrate side, and FIG. **7B** is a top-emission lighting device in which light is extracted from the sealing substrate side.

(156) A lighting device **4000** illustrated in FIG. **7A** includes a light-emitting device **4002** over a substrate **4001**. In addition, the lighting device **4000** includes a substrate **4003** with unevenness on the outside of the substrate **4001**. The light-emitting device **4002** includes a first electrode **4004**, an EL layer **4005**, and a second electrode **4006**.

(157) The first electrode **4004** is electrically connected to an electrode **4007**, and the second electrode **4006** is electrically connected to an electrode **4008**. In addition, an auxiliary wiring **4009**

electrically connected to the first electrode **4004** may be provided. Note that an insulating layer **4010** is formed over the auxiliary wiring **4009**.

(158) The substrate **4001** and a sealing substrate **4011** are bonded to each other with a sealant **4012**. A desiccant **4013** is preferably provided between the sealing substrate **4011** and the light-emitting device **4002**. The substrate **4003** has the unevenness illustrated in FIG. 7A, whereby the extraction efficiency of light generated in the light-emitting device **4002** can be increased.

(159) A lighting device **4200** illustrated in FIG. 7B includes a light-emitting device **4202** over a substrate **4201**. The light-emitting device **4202** includes a first electrode **4204**, an EL layer **4205**, and a second electrode **4206**.

(160) The first electrode **4204** is electrically connected to an electrode **4207**, and the second electrode **4206** is electrically connected to an electrode **4208**. An auxiliary wiring **4209** electrically connected to the second electrode **4206** may also be provided. In addition, an insulating layer **4210** may be provided under the auxiliary wiring **4209**.

(161) The substrate **4201** and a sealing substrate **4211** with unevenness are bonded to each other with a sealant **4212**. A barrier film **4213** and a planarization film **4214** may be provided between the sealing substrate **4211** and the light-emitting device **4202**. The sealing substrate **4211** has the unevenness illustrated in FIG. 7B, whereby the extraction efficiency of light generated in the light-emitting device **4202** can be increased.

(162) Application examples of such lighting devices include a ceiling light for indoor lighting. Examples of the ceiling light include a ceiling direct mount light and a ceiling embedded light. Such a lighting device is fabricated using the light-emitting apparatus in combination with a housing or a cover.

(163) For another example, such lighting devices can be used for a foot light that illuminates a floor so that the safety of one's feet can be improved. For example, the foot light can be effectively used in a bedroom, on a staircase, or on a passage. In that case, the size or shape of the foot light can be changed depending on the area or structure of a room. The foot light can be a stationary lighting device fabricated using the light-emitting apparatus in combination with a support base.

(164) Such lighting devices can also be used for a sheet-like lighting device (sheet-like lighting). The sheet-like lighting, which is attached to a wall when used, is space-saving and thus can be used for a wide variety of uses. Furthermore, the area of the sheet-like lighting can be easily increased. The sheet-like lighting can also be used on a wall or housing having a curved surface.

(165) Besides the above examples, the light-emitting apparatus which is one embodiment of the present invention or the light-emitting device which is a part of the light-emitting apparatus can be used as part of furniture in a room, so that a lighting device which has a function of the furniture can be obtained.

(166) As described above, a variety of lighting devices that include the light-emitting apparatus can be obtained. Note that these lighting devices are also included in embodiments of the present invention.

(167) The structure described in this embodiment can be used in an appropriate combination with any of the structures described in the other embodiments.

Example 1

(168) In this example, three kinds of light-emitting devices having different stacked-layer structures of an EL layer **902**, including the light-emitting device of one embodiment of the present invention, are fabricated and the obtained device characteristics are shown. Note that a light-emitting device **1** fabricated in this example has a structure in which an electron-transport layer **914** formed using an electron-transport material and an electron-injection layer **915** formed using an alkali metal compound are stacked between a light-emitting layer **913** and a second electrode **903**. A light-emitting device **2** includes, between the light-emitting layer **913** and the second electrode **903**, the electron-transport layer **914** formed using an alkali metal and an organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton, which is

used for the light-emitting layer **913**. A comparative light-emitting device **3** includes, between the light-emitting layer **913** and the second electrode **903**, a three-layer structure: the electron-transport layer **914** having a two-layer structure and an electron-injection layer **915**.

(169) Specific device structures and fabrication methods of the above light-emitting devices will be described below. Note that FIG. **8** illustrates a device structure of the light-emitting devices described in this example, and Table 1 shows specific structures. Chemical formulae of materials used in this example are shown below.

(170) TABLE-US-00001 TABLE 1 Hole- Hole- Light- Electron- First injection transport emitting injection Second electrode layer layer layer Electron-transport layer layer electrode 901 911 912 913 914 915 903 Light- ITSO HAT-CN PCBBiF * — NBphen LiF Al emitting (70 nm) (5 nm) (70 nm) (70 nm) (15 nm) (1 nm) (200 nm) device 1 Light- ITSO HAT-CN PCBBiF ** 9mDBtBPNfpr:LiQ — Al emitting (70 nm) (5 nm) (70 nm) (80 nm) (1:1 5 nm) (200 nm) device 2 Comparative ITSO HAT-CN PCBBiF *** 9mDBtBPNfpr NBphen LiF Al light- (70 nm) (5 nm) (70 nm) (40 nm) 30 nm) (15 nm) (1 nm) (200 nm) emitting device 3 * 9mDBtBPNfpr: PCBBiF: [Ir(dmpqn).sub.2(acac)] (0.7:0.3:0.1) ** 9mDBtBPNfpr: PCBBiF: [Ir(dmpqn).sub.2(acac)] (0.8:0.2:0.1) *** 9mDBtBPNfpr: PCBBiF: [Ir(dmpqn).sub.2(acac)] (0.75:0.25:0.1)

(171) ##STR00017## ##STR00018##

<<Fabrication of Light-Emitting Devices>>

<Fabrication of Light-Emitting Device **1**, Light-Emitting Device **2**, and Comparative Light-Emitting Device **3**>

(172) The light-emitting devices described in this example each have a structure, as illustrated in FIG. **8**, in which a hole-injection layer **911**, a hole-transport layer **912**, the light-emitting layer **913**, the electron-transport layer **914**, and the electron-injection layer **915** are stacked in this order over a first electrode **901** formed over a substrate **900**, and the second electrode **903** is stacked over the electron-injection layer **915**.

(173) First, the first electrode **901** was formed over the substrate **900**. The electrode area was set to 4 mm.sup.2 (2 mm×2 mm). A glass substrate was used as the substrate **900**. The first electrode **901** was formed by deposition of indium tin oxide containing silicon oxide (ITSO) by a sputtering method to a thickness of 70 nm.

(174) As pretreatment, a surface of the substrate was washed with water, baking was performed at 200° C. for one hour, and then UV ozone treatment was performed for 370 seconds. After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to approximately 10.sup.-4 Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was cooled down for approximately 30 minutes.

(175) Next, the hole-injection layer **911** was formed over the first electrode **901**. The hole-injection layer **911** was formed by evaporation using HAT-CN to a thickness of 5 nm.

(176) Then, the hole-transport layer **912** was formed over the hole-injection layer **911**. The hole-transport layer **912** was formed by evaporation using N-(1,1'-biphenyl-4-yl)-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9-dimethyl-9H-fluoren-2-amine (abbreviation: PCBBiF) to a thickness of 70 nm.

(177) Next, the light-emitting layer **913** was formed over the hole-transport layer **912**.

(178) In the case of the light-emitting device **1**, the light-emitting layer **913** was formed by co-evaporation using bis[4,6-dimethyl-2-(2-quinolinyl-κN)phenyl-κC](2,4-pentanedionato-κ.sup.2O,O')iridium(III) (abbreviation: [Ir(dmpqn).sub.2(acac)]) as a guest material (a phosphorescent material) in addition to 9-[3'-(dibenzothiophen-4-yl)bipheny-3-yl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr) and PCBBiF so that a weight ratio was 9mDBtBPNfpr: PCBBiF: [Ir(dmpqn).sub.2(acac)]=0.7:0.3:0.1. The thickness was set to 70 nm.

(179) In the case of the light-emitting device **2**, co-evaporation was performed using materials

similar to those for the light-emitting device **1** so that the weight ratio was 9mDBtBPNfpr:

PCBBiF: [Ir(dmpqn).sub.2(acac)]=0.8:0.2:0.1. The thickness was set to 80 nm.

(180) In the case of the comparative light-emitting device **3**, co-evaporation was performed using a material similar to that for the light-emitting device **1** so that the weight ratio was 9mDBtBPNfpr:

PCBBiF: [Ir(dmpqn).sub.2(acac)]=0.75:0.25:0.1. The thickness was set to 40 nm.

(181) Next, the electron-transport layer **914** was formed over the light-emitting layer **913**.

(182) In the case of the light-emitting device **1**, the electron-transport layer **914** was formed to a thickness of 15 nm by evaporation using 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBphen) that is an electron-transport material. In the case of the light-emitting device **2**, co-evaporation was performed using 8-quinolinolato-lithium (abbreviation: Liq) and 9mDBtBPNfpr that was the same electron-transport material as that used for the light-emitting layer **913** so that the weight ratio was 9mDBtBPNfpr: Liq=1:1. The thickness was set to 5 nm. In the case of the comparative light-emitting device **3**, evaporation was performed using 9mDBtBPNfpr that was the same electron-transport material as that used for the light-emitting layer **913** so that the thickness was 30 nm, and then evaporation was performed using NBphen that was an electron-transport material so that the thickness was 15 nm.

(183) Then, the electron-injection layer **915** was formed over the electron-transport layer **914**. Note that the electron-injection layer **915** was formed only in the light-emitting device **1** and the comparative light-emitting device **3**. The electron-injection layer **915** was formed by evaporation using lithium fluoride (LiF) to a thickness of 1 nm.

(184) After that, the second electrode **903** was formed over the electron-injection layer **915**. The second electrode **903** was formed by an evaporation method using aluminum to a thickness of 200 nm. In this example, the second electrode **903** functions as a cathode.

(185) Through the above steps, the light-emitting devices in each of which an EL layer was sandwiched between a pair of electrodes over the substrate **900** were formed. The hole-injection layer **911**, the hole-transport layer **912**, the light-emitting layer **913**, the electron-transport layer **914**, and the electron-injection layer **915** described in the above steps are functional layers forming the EL layer in one embodiment of the present invention. Furthermore, in all the evaporation steps in the above fabrication method, an evaporation method by a resistance-heating method was used.

(186) The light-emitting devices fabricated as described above were sealed using another substrate (not illustrated). At the time of the sealing using the another substrate (not illustrated), the another substrate (not illustrated) on which a sealant that solidifies by ultraviolet light was applied was fixed onto the substrate **900** in a glove box containing a nitrogen atmosphere, and the substrates were bonded to each other such that the sealant attached to the periphery of the light-emitting device formed over the substrate **900**. At the time of the sealing, the sealant was irradiated with 365-nm ultraviolet light at 6 J/cm.sup.2 to be solidified, and the sealant was subjected to heat treatment at 80° C. for one hour to be stabilized.

(187) <<Operation Characteristics of Light-Emitting Devices>>

(188) Operation characteristics of each of the fabricated light-emitting devices were measured. Note that the measurement was carried out at room temperature (an atmosphere maintained at 25° C.). The results are shown in FIG. **9** to FIG. **12**.

(189) Table 2 below shows initial values of main characteristics of the light-emitting devices at around 1000 cd/m.sup.2.

(190) TABLE-US-00002

	TABLE 2	External	Current	Current	Power	quantum	Voltage	Current
		density	Chromaticity	Luminance	efficiency	efficiency	(V)	(mA)
		(cd/m.sup.2)	(cd/A)	(lm/W)	(%)	Light-	3.3	0.19
						4.7	(0.68,	0.32)
						990	21	20
						22	emitting	device 1
						Light-	3.7	0.20
						5.0	(0.68,	0.32)
						1000	21	17
						22	emitting	device 2
						Comparative	3.2	0.18
						4.4	(0.68,	0.32)
						950	22	21
						23	light-	emitting
						device 3		

(191) The above results show that there is no significant difference in initial characteristics between the light-emitting devices described in this example despite their different device structures (e.g.,

stacked-layer structures and thicknesses).

(192) FIG. 13 shows emission spectra when current at a current density of 2.5 mA/cm² was applied to the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3. As shown in FIG. 13, the emission spectra of the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3 each have a peak at around 624 nm, indicating that the peak is derived from light emission of [Ir(dmpqn).sub.2(acac)] contained in the light-emitting layer 913.

(193) Next, reliability tests were performed on the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3. FIG. 14 shows the results of the reliability tests. In FIG. 14, the vertical axis represents normalized luminance (%) with an initial luminance of 100%, and the horizontal axis represents driving time (h) of the device. As the reliability tests, constant current driving tests where a constant current was supplied at a current density of 75 mA/cm² were performed.

(194) The results of the reliability tests show that the light-emitting device 1 and the light-emitting device 2 have no significant difference from the comparative light-emitting device 3.

(195) The comparative light-emitting device 3 has a structure with improved reliability by having, between the light-emitting layer 913 and the second electrode 903, a stacked-layer structure of three layers: the electron-transport layer 914 in which a first layer containing 9mDBtBPNfpr used as a host material in the light-emitting layer 913 and a second layer containing NBphen are stacked, and the electron-injection layer 915 formed using LiF; whereas the light-emitting device 1 has a two-layer structure in which the electron-transport layer 914 formed using only NBphen and the electron-injection layer 915 formed using LiF are stacked. The light-emitting device 2 has a single-layer structure between the light-emitting layer 913 and the second electrode 903 with only the electron-transport layer 914 formed using a mixed film of Liq and 9mDBtBPNfpr that is used as the host material in the light-emitting layer 913.

(196) The light-emitting devices fabricated in this example are each characterized in that an electron-transport organic compound having a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton is used as a host material in the light-emitting layer 913.

(197) The results described in this example show that, although the light-emitting device 1, the light-emitting device 2, and the comparative light-emitting device 3 have different stacked-layer structure, reliability similar to that of the comparative light-emitting device 3 with a larger number of stacked layers can be obtained. Thus, the light-emitting device 1 and the light-emitting device 2 can each be fabricated through simpler process than the comparative light-emitting device 3.

Furthermore, thickness adjustment in optical design can be performed with the light-emitting layer 913, so that a light-emitting device that does not easily deteriorate can be provided by increasing the thickness of the light-emitting layer 913.

Reference Synthesis Example 1

(198) A synthesis method of 9-[(3'-dibenzothiophen-4-yl)biphenyl-3-yl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr), which is the organic compound used in this example and represented by Structural Formula (100), is described. The structure of 9mDBtBPNfpr is shown below.

(199) ##STR00019##

Step 1; Synthesis of 6-chloro-3-(2-methoxynaphthalen-1-yl)pyrazin-2-amine

(200) First, into a three-neck flask equipped with a reflux pipe were put 4.37 g of 3-bromo-6-chloropyrazin-2-amine, 4.23 g of 2-methoxynaphthalene-1-boronic acid, 4.14 g of potassium fluoride, and 75 mL of dehydrated tetrahydrofuran, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.57 g of tris(dibenzylideneacetone)dipalladium(0) (abbreviation: Pd.sub.2(dba).sub.3) and 4.5 mL of tri-tert-butylphosphine (abbreviation: P(tBu).sub.3) were added thereto, and then stirring was performed at 80° C. for 54 hours for reaction.

(201) After a predetermined time elapsed, the obtained mixture was subjected to suction filtration and the filtrate was concentrated. Then, purification by silica gel column chromatography using a developing solvent of toluene: ethyl acetate=9:1 was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 2.19 g, in a yield of 36%). The synthesis scheme of Step 1 is shown in Formula (a-1) below.

(202) ##STR00020##

Step 2; Synthesis of 9-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine

(203) Next, into a three-neck flask were put 2.18 g of 6-chloro-3-(2-methoxynaphthalen-1-yl)pyrazin-2-amine obtained in Step 1, 63 mL of dehydrated tetrahydrofuran, and 84 mL of a glacial acetic acid, and the air in the flask was replaced with nitrogen. After the flask was cooled down to -10° C., 2.8 mL of tert-butyl nitrite was dripped, and stirring was performed at -10° C. for 30 minutes and at 0° C. for 3 hours. After a predetermined time elapsed, 250 mL of water was added to the obtained suspension and suction filtration was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 1.48 g, in a yield of 77%). The synthesis scheme of Step 2 is shown in (a-2) below.

(204) ##STR00021##

Step 3; Synthesis of 9-[(3'-dibenzothiophene-4-yl)biphenyl-3-yl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr)

(205) Into a three-neck flask were put 1.48 g of 9-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine obtained in Step 2, 3.41 g of 3'-(4-dibenzothiophene)-1,1'-biphenyl-3-boronic acid, 8.8 mL of a 2M aqueous solution of potassium carbonate, 100 mL of toluene, and 10 mL of ethanol, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.84 g of bis(triphenylphosphine)palladium(II) dichloride (abbreviation: Pd(PPh.sub.3).sub.2Cl.sub.2) was added thereto, and then stirring was performed at 80° C. for 18 hours for reaction.

(206) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was dissolved in toluene, and the mixture was filtered through a filter aid in which Celite, alumina, and Celite were stacked in this order and was recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (a pale yellow solid, 2.66 g, in a yield of 82%).

(207) By a train sublimation method, 2.64 g of the obtained pale yellow solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.6 Pa at 315° C. while the argon gas flowed at a flow rate of 15 mL/min. After the purification by sublimation, 2.34 g of a target pale yellow solid was obtained in a yield of 89%. The synthesis scheme of Step 3 is shown in (a-3) below.

(208) ##STR00022##

(209) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the pale yellow solid obtained in Step 3 are shown below.

(210) .sup.1H-NMR. δ (CD.sub.2Cl.sub.2): 7.47-7.51 (m, 2H), 7.60-7.69 (m, 5H), 7.79-7.89 (m, 6H), 8.05 (d, 1H), 8.10-8.11 (m, 2H), 8.18-8.23 (m, 3H), 8.53 (s, 1H), 9.16 (d, 1H), 9.32 (s, 1H).
Reference Synthesis Example 2

(211) A synthesis method of 9-(9'-phenyl-3,3'-bi-9H-carbazol-9-yl)naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9PCCzNfpr), which is the organic compound represented by Structural Formula (101) in Embodiment 1, is described. The structure of 9PCCzNfpr is shown below.

(212) ##STR00023##

(213) Into a three-neck flask were put 0.94 g of 9-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 2 in Reference Synthesis Example 1, 1.69 g of 9'-phenyl-3,3'-bi-9H-carbazole, and 37 mL of mesitylene, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 1.23 g of sodium tert-butoxide, 0.021 g of tris(dibenzylideneacetone)dipalladium(0) (abbreviation:

Pd.sub.2(dba).sub.3), and 0.030 g of 2-dicyclohexylphosphino-2',6'-dimethoxybiphenyl (abbreviation: S-Phos) were added thereto, and then stirring was performed at 120° C. for 8 hours for reaction.

(214) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was dissolved in toluene, and the mixture was filtered through a filter aid in which Celite, alumina, and Celite were stacked in this order and was recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (a yellow solid, 0.85 g, in a yield of 36%).

(215) By a train sublimation method, 0.84 g of the obtained yellow solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.5 Pa at 350° C. while the argon gas flowed at a flow rate of 10 mL/min. After the purification by sublimation, 0.64 g of a target yellow solid was obtained in a yield of 76%. The synthesis scheme of the above synthesis method is shown in (b-1) below.

(216) ##STR00024##

(217) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the yellow solid obtained by the above synthesis method are shown below.

(218) .sup.1H-NMR. δ (CDCl.sub.3): 7.32-7.35 (m, 1H), 7.42-7.57 (m, 6H), 7.63-7.70 (m, 5H), 7.80-7.90 (m, 4H), 8.09 (d, 2H), 8.14 (d, 2H), 8.27 (d, 2H), 8.49 (d, 2H), 9.20 (d, 1H), 9.27 (s, 1H).

Reference Synthesis Example 3

(219) A synthesis method of 9-[3-(9'-phenyl-3,3'-bi-9H-carbazol-9-yl)phenyl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mPCCzPNfpr), which is the organic compound represented by Structural Formula (102) in Embodiment 1, is described. The structure of 9mPCCzPNfpr is shown below.

(220) ##STR00025##

Step 1; Synthesis of 9-(3-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine

(221) Into a three-neck flask were put 2.12 g of 9-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 2 in Reference Synthesis Example 1, 1.41 g of 3-chlorophenylboronic acid, 14 mL of a 2M aqueous solution of potassium carbonate, 83 mL of toluene, and 8.3 mL of ethanol, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.19 g of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 1.12 g of tris(2,6-dimethoxyphenyl)phosphine (abbreviation: P(2,6-MeOPh).sub.3) were added thereto, and then stirring was performed at 90° C. for 7.5 hours for reaction.

(222) After a predetermined time elapsed, the obtained mixture was subjected to suction filtration, followed by washing with ethanol. Then, purification by silica gel column chromatography using toluene as a developing solvent was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 1.97 g, in a yield of 73%). The synthesis scheme of Step 1 is shown in Formula (c-1) below.

(223) ##STR00026##

Step 2; Synthesis of 9mPCCzPNfpr

(224) Next, into a three-neck flask were put 1.45 g of 9-(3-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine obtained in Step 1, 1.82 g of 9'-phenyl-3,3'-bi-9H-carbazole, and 22 mL of mesitylene, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.85 g of sodium tert-butoxide, 0.025 g of tris(dibenzylideneacetone)dipalladium(0) (abbreviation: Pd.sub.2(dba).sub.3), and 0.036 g of 2-dicyclohexylphosphino-2',6'-dimethoxybiphenyl (abbreviation: S-Phos) were added thereto, and then stirring was performed at 150° C. for 7 hours for reaction.

(225) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was dissolved in toluene, and the mixture was filtered through a filter aid in which Celite, alumina, and Celite were

stacked in this order and was recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (a yellow solid, 2.22 g, in a yield of 71%).

(226) By a train sublimation method, 2.16 g of the obtained yellow solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.6 Pa at 385° C. while the argon gas flowed at a flow rate of 18 mL/min. After the purification by sublimation, 1.67 g of a target yellow solid was obtained in a yield of 77%. The synthesis scheme of Step 2 is shown in Formula (c-2) below.

(227) ##STR00027##

(228) Results of analysis by nuclear magnetic resonance (¹H-NMR) spectroscopy of the yellow solid obtained in Step 2 are shown below.

(229) ¹H-NMR. δ (CDCl₃): 7.31-7.39 (m, 2H), 7.43-7.59 (m, 6H), 7.64-7.69 (m, 6H), 7.78-7.88 (m, 6H), 8.09 (d, 1H), 8.15 (d, 1H), 8.26 (d, 1H), 8.30 (d, 1H), 8.34 (d, 1H), 8.51-8.55 (m, 3H), 9.15 (d, 1H), 9.35 (s, 1H).

Reference Synthesis Example 4

(230) A synthesis method of 9-[3-(9'-phenyl-2,3'-bi-9H-carbazol-9-yl)phenyl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mPCCzPNfpr-02), which is the organic compound represented by Structural Formula (103) in Embodiment 1, is described. The structure of 9mPCCzPNfpr-02 is shown below.

(231) ##STR00028##

(232) Into a three-neck flask were put 1.19 g of 9-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 2 in Reference Synthesis Example 1, 3.51 g of 3-(9'-phenyl-2,3'-bi-9H-carbazol-9-yl)phenylboronic acid pinacol ester, 6.0 mL of a 2M aqueous solution of potassium carbonate, 60 mL of toluene, and 6 mL of ethanol, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.33 g of bis(triphenylphosphine)palladium(II) dichloride (abbreviation: Pd(PPh₃)₂Cl₂) was added thereto, and then stirring was performed at 90° C. for 16 hours for reaction.

(233) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was dissolved in toluene, and the mixture was filtered through a filter aid in which Celite, alumina, and Celite were stacked in this order and was recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (a yellow solid, 3.01 g, in a yield of 90%).

(234) By a train sublimation method, 3.00 g of the obtained yellow solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.7 Pa at 380° C. while the argon gas flowed at a flow rate of 16 mL/min. After the purification by sublimation, 2.47 g of a target yellow solid was obtained in a yield of 82%. The synthesis scheme is shown in Formula (d-1) below.

(235) ##STR00029##

(236) Results of analysis by nuclear magnetic resonance (¹H-NMR) spectroscopy of the yellow solid obtained above are shown below.

(237) ¹H-NMR. δ (CDCl₃): 7.22-7.25 (m, 1H), 7.34-7.42 (m, 3H), 7.46-7.49 (m, 3H), 7.55-7.66 (m, 6H), 7.72-7.88 (m, 7H), 8.07 (d, 1H), 8.13 (d, 1H), 8.19-8.22 (m, 2H), 8.28 (d, 1H), 8.33 (d, 1H), 8.46 (s, 1H), 8.54 (s, 1H), 9.14 (d, 1H), 9.34 (s, 1H).

Reference Synthesis Example 5

(238) A synthesis method of 10-[(3'-dibenzothiophen-4-yl)bipheny-3-yl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 10mDBtBPNfpr), which is the organic compound represented by Structural Formula (104) in Embodiment 1, is described. The structure of 10mDBtBPNfpr is shown below.

(239) ##STR00030##

Step 1; Synthesis of 5-chloro-3-(2-methoxynaphthalen-1-yl)pyrazin-2-amine

(240) First, into a three-neck flask equipped with a reflux pipe were put 5.01 g of 3-bromo-5-chloropyrazin-2-amine, 6.04 g of 2-methoxynaphthalene-1-boronic acid, 5.32 g of potassium fluoride, and 86 mL of dehydrated tetrahydrofuran, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.44 g of tris(dibenzylideneacetone)dipalladium(0) (abbreviation: Pd.sub.2(dba).sub.3) and 3.4 mL of tri-tert-butylphosphine (abbreviation: P(tBu).sub.3) were added thereto, and then stirring was performed at 80° C. for 22 hours for reaction.

(241) After a predetermined time elapsed, the obtained mixture was subjected to suction filtration and the filtrate was concentrated. Then, purification by silica gel column chromatography using a developing solvent of toluene: ethyl acetate=10:1 was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 5.69 g, in a yield of 83%). The synthesis scheme of Step 1 is shown in Formula (e-1) below.

(242) ##STR00031##

Step 2; Synthesis of 10-chloro-naphtho[1',2':4,5]furo[2,3-b]pyrazine

(243) Next, into a three-neck flask were put 5.69 g of 5-chloro-3-(2-methoxynaphthalen-1-yl)pyrazin-2-amine obtained in Step 1, 150 mL of dehydrated tetrahydrofuran, and 150 mL of a glacial acetic acid, and the air in the flask was replaced with nitrogen. After the flask was cooled down to -10° C., 7.1 mL of tert-butyl nitrite was dripped, and stirring was performed at -10° C. for 1 hour and at 0° C. for 3.5 hours. After a predetermined time elapsed, 1 L of water was added to the obtained suspension and suction filtration was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 4.06 g, in a yield of 81%). The synthesis scheme of Step 2 is shown in Formula (e-2) below.

(244) ##STR00032##

Step 3; Synthesis of 10mDBtBPNfpr

(245) Into a three-neck flask were put 1.18 g of 10-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine obtained in Step 2, 2.75 g of 3'-(4-dibenzothiophene)-1,1'-biphenyl-3-boronic acid, 7.5 mL of a 2M aqueous solution of potassium carbonate, 60 mL of toluene, and 6 mL of ethanol, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.66 g of bis(triphenylphosphine)palladium(II) dichloride (abbreviation: Pd(PPh.sub.3).sub.2Cl.sub.2) was added thereto, and then stirring was performed at 90° C. for 22.5 hours for reaction.

(246) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was dissolved in toluene, and the mixture was filtered through a filter aid in which Celite, alumina, and Celite were stacked in this order and was recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (a white solid, 2.27 g, in a yield of 87%).

(247) By a train sublimation method, 2.24 g of the obtained white solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.3 Pa at 310° C. while the argon gas flowed at a flow rate of 16 mL/min. After the purification by sublimation, 1.69 g of a target white solid was obtained in a yield of 75%. The synthesis scheme of Step 3 is shown in Formula (e-3) below.

(248) ##STR00033##

(249) Results of analysis by nuclear magnetic resonance spectroscopy (.sup.1H-NMR) of the white solid obtained in Step 3 are shown below.

(250) .sup.1H-NMR. δ (CDCl.sub.3): 7.43 (t, 1H), 7.48 (t, 1H), 7.59-7.62 (m, 3H), 7.68-7.86 (m, 8H), 8.05 (d, 1H), 8.12 (d, 1H), 8.18 (s, 1H), 8.20-8.24 (m, 3H), 8.55 (s, 1H), 8.92 (s, 1H), 9.31 (d, 1H).

Reference Synthesis Example 6

(251) A synthesis method of 10-(9'-phenyl-3,3'-bi-9H-carbazol-9-yl)naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 10PCCzNfpr), which is the organic compound represented by Structural

Formula (105) in Embodiment 1, is described. The structure of 10PCCzNfpr is shown below.

(252) ##STR00034##

(253) Into a three-neck flask were put 1.80 g of 10-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 2 in Reference Synthesis Example 5, 3.10 g of 9'-phenyl-3,3'-bi-9H-carbazole, and 71 mL of mesitylene, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 2.21 g of sodium tert-butoxide, 0.041 g of tris(dibenzylideneacetone)dipalladium(0) (abbreviation: Pd.sub.2(dba).sub.3), and 0.061 g of 2-dicyclohexylphosphino-2',6'-dimethoxybiphenyl (abbreviation: S-Phos) were added thereto, and then stirring was performed at 120° C. for 2 hours for reaction.

(254) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was dissolved in toluene, and the mixture was filtered through a filter aid in which Celite, alumina, and Celite were stacked in this order and was recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (an orange solid, 3.47 g, in a yield of 78%).

(255) By a train sublimation method, 3.42 g of the obtained orange solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.4 Pa at 350° C. while the argon gas flowed at a flow rate of 16 mL/min. After the purification by sublimation, 2.86 g of a target orange solid was obtained in a yield of 84%. The synthesis scheme is shown in (f-1) below.

(256) ##STR00035##

(257) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the orange solid obtained by the above synthesis method are shown below.

(258) .sup.1H-NMR. δ (CDCl.sub.3): 7.32-7.35 (m, 1H), 7.43-7.57 (m, 6H), 7.63-7.68 (m, 5H), 7.79-7.84 (m, 2H), 7.89-7.91 (m, 2H), 8.01 (d, 1H), 8.07-8.09 (m, 2H), 8.18 (d, 1H), 8.27 (d, 1H), 8.30 (d, 1H), 8.51 (s, 2H), 8.85 (s, 1H), 9.16 (d, 1H).

Reference Synthesis Example 7

(259) A synthesis method of 12-[(3'-dibenzothiophen-4-yl)biphenyl-3-yl]phenanthro[9',10':4,5]furo[2,3-b]pyrazine (abbreviation: 12mDBtBPPnfpr), which is the organic compound represented by Structural Formula (106) in Embodiment 1, is described. The structure of 12mDBtBPPnfpr is shown below.

(260) ##STR00036##

Step 1; Synthesis of 9-methoxyphenanthrene

(261) First, into a three-neck flask equipped with a reflux pipe were put 4.02 g of 9-bromo-phenanthrene, 7.80 g of cesium carbonate, 16 mL of toluene, and 16 mL of methanol, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.11 g of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 0.41 g of 2-di-tert-butylphosphino-2',4',6'-triisopropylbiphenyl (abbreviation: tBuXPhos) were added thereto, and then stirring was performed at 80° C. for 17 hours for reaction.

(262) After a predetermined time elapsed, the obtained mixture was subjected to suction filtration and the filtrate was concentrated. Then, purification by silica gel column chromatography using a developing solvent of toluene: hexane=1:3 was performed, whereby a target substance was obtained (white powder, 2.41 g, in a yield of 74%). The synthesis scheme of Step 1 is shown in Formula (g-1) below.

(263) ##STR00037##

Step 2; Synthesis of 9-bromo-10-methoxyphenanthrene

(264) Next, into a conical flask were put 2.75 g of 9-methoxyphenanthrene obtained in Step 1, 0.18 mL of diisopropylamine, 150 mL of dehydrated dichloromethane, and 2.52 g of N-bromosuccinimide (abbreviation: NBS), and stirring was performed at room temperature for 18 hours. After a predetermined time elapsed, the mixture was washed with water and an aqueous

solution of sodium thiosulfate, and then concentrated. After that, purification by silica gel column chromatography using a developing solvent of hexane:ethyl acetate=5:1 was performed, whereby a target substance was obtained (yellowish white powder, 2.46 g, in a yield of 65%). The synthesis scheme of Step 2 is shown in Formula (g-2) below.

(265) ##STR00038##

Step 3; Synthesis of 10-methoxyphenanthrene-9-boronic acid

(266) Next, into a three-neck flask were put 8.49 g of 9-bromo-10-methoxyphenanthrene obtained in Step 2 and 250 mL of dehydrated THF, and the air in the flask was replaced with nitrogen. After the flask was cooled down to -78°C ., 22 mL of a 1.6M hexane solution of n-butyllithium was added, and stirring was performed at -78°C . for 3 hours. Then, 5.7 mL of tetramethylethylenediamine and 4.3 mL of trimethyl borate were added, and stirring was performed at room temperature for 18 hours for reaction.

(267) After a predetermined time elapsed, 50 mL of 1M hydrochloric acid was added, and stirring was performed at room temperature for 1 hour. Then, extraction with toluene was performed, whereby a target substance was obtained (pale orange powder, 2.87 g, in a yield of 39%). The synthesis scheme of Step 3 is shown in Formula (g-3) below.

(268) ##STR00039##

Step 4; Synthesis of 5-chloro-3-(10-methoxyphenanthren-9-yl)pyrazin-2-amine

(269) Next, into a three-neck flask equipped with a reflux pipe were put 3.69 g of 10-methoxyphenanthrene-9-boronic acid obtained in Step 3, 3.02 g of 3-bromo-5-chloropyrazin-2-amine, 70 mL of toluene, and 35 mL of a 2M sodium carbonate aqueous solution, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.16 g of tetrakis(triphenylphosphine)palladium(0) (abbreviation: Pd(PPh.sub.3).sub.4) was added thereto, and then stirring was performed at 110°C . for 7.5 hours for reaction.

(270) After a predetermined time elapsed, extraction with toluene was performed. Then, purification by flash column chromatography using a developing solvent of dichloromethane: ethyl acetate=50:1 was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 3.00 g, in a yield of 62%). The synthesis scheme of Step 4 is shown in Formula (g-4) below.

(271) ##STR00040##

Step 5; Synthesis of 12-chlorophenanthro[9',10':4,5]furo[2,3-b]pyrazine

(272) Next, into a three-neck flask were put 2.92 g of 5-chloro-3-(10-methoxyphenanthren-9-yl)pyrazin-2-amine obtained in Step 4, 60 mL of dehydrated tetrahydrofuran, and 60 mL of a glacial acetic acid, and the air in the flask was replaced with nitrogen. After the flask was cooled down to -10°C ., 3.1 mL of tert-butyl nitrite was dripped, and stirring was performed at -10°C . for 1 hour and at 0°C . for 22 hours.

(273) After a predetermined time elapsed, 200 mL of water was added to the obtained suspension and suction filtration was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 2.06 g, in a yield of 80%). The synthesis scheme of Step 5 is shown in (g-5) below.

(274) ##STR00041##

Step 6; Synthesis of 12-(3-chlorophenyl)phenanthro[9',10':4,5]furo[2,3-b]pyrazine

(275) Next, into a three-neck flask were put 1.02 g of 12-chlorophenanthro[9',10':4,5]furo[2,3-b]pyrazine obtained in Step 5, 0.56 g of 3-chlorophenylboronic acid, 5 mL of a 2M aqueous solution of potassium carbonate, 33 mL of toluene, and 3.3 mL of ethanol, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.074 g of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 0.44 g of tris(2,6-dimethoxyphenyl)phosphine (abbreviation: P(2,6-MeOPh).sub.3) were added thereto, and then stirring was performed at 90°C . for 5.5 hours for reaction.

(276) After a predetermined time elapsed, the obtained mixture was subjected to suction filtration

and the filtrate was concentrated. Then, purification by silica gel column chromatography using toluene as a developing solvent was performed, whereby a target pyrazine derivative was obtained (white powder, 0.87 g, in a yield of 70%). The synthesis scheme of Step 6 is shown in Formula (g-6) below.

(277) ##STR00042##

Step 7; Synthesis of 12mDBtBPPnfpr

(278) Next, into a three-neck flask were put 0.85 g of 12-(3-chlorophenyl)phenanthro[9',10':4,5]furo[2,3-b]pyrazine obtained in Step 6, 0.73 g of 3-(4-dibenzothiophene)phenylboronic acid, 1.41 g of tripotassium phosphate, 0.49 g of tert-butyl alcohol, and 18 mL of diethylene glycol dimethyl ether (abbreviation: diglyme), and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 9.8 mg of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 32 mg of di(1-adamantyl)-n-butylphosphine (abbreviation: CataCXium A) were added thereto, and then stirring was performed at 140° C. for 11.5 hours for reaction.

(279) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was dissolved in toluene, and the mixture was filtered through a filter aid in which Celite, alumina, and Celite were stacked in this order and was recrystallized with toluene, whereby a target substance was obtained (a white solid, 0.74 g, in a yield of 55%).

(280) By a train sublimation method, 0.73 g of the obtained white solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.6 Pa at 330° C. while the argon gas flowed at a flow rate of 11 mL/min. After the purification by sublimation, 0.49 g of a target white solid was obtained in a yield of 67%. The synthesis scheme of Step 7 is shown in Formula (g-7) below.

(281) ##STR00043##

(282) Results of analysis by nuclear magnetic resonance spectroscopy (.sup.1H-NMR) of the white solid obtained in Step 7 are shown below.

(283) .sup.1H-NMR. δ (CD.sub.2Cl.sub.2): 7.45 (t, 1H), 7.50 (t, 1H), 7.62-7.66 (m, 2H), 7.70-7.89 (m, 10H), 8.21-8.28 (m, 4H), 8.58-8.61 (m, 2H), 8.80 (d, 1H), 8.84 (d, 1H), 8.94 (s, 1H), 9.37 (d, 1H).

Reference Synthesis Example 8

(284) A synthesis method of 9-[4-(9'-phenyl-3,3'-bi-9H-carbazol-9-yl)phenyl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9pPCCzPNfpr), which is the organic compound represented by Structural Formula (107) in Embodiment 1, is described. The structure of 9pPCCzPNfpr is shown below.

(285) ##STR00044##

Step 1; Synthesis of 9-(4-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine

(286) Into a three-neck flask were put 4.10 g of 9-chloronaphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 2 in Reference Synthesis Example 1, 2.80 g of 4-chlorophenylboronic acid, 27 mL of a 2M aqueous solution of potassium carbonate, 160 mL of toluene, and 16 mL of ethanol, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.36 g of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 2.08 g of tris(2,6-dimethoxyphenyl)phosphine (abbreviation: P(2,6-MeOPh).sub.3) were added thereto, and then stirring was performed at 90° C. for 7 hours for reaction.

(287) After a predetermined time elapsed, the obtained mixture was subjected to suction filtration, followed by washing with ethanol. Then, purification by silica gel column chromatography using toluene as a developing solvent was performed, whereby a target pyrazine derivative was obtained (yellowish white powder, 2.81 g, in a yield of 52%). The synthesis scheme of Step 1 is shown in Formula (h-1) below.

(288) ##STR00045##

Step 2; Synthesis of 9pPCCzPNfpr

(289) Next, into a three-neck flask were put 1.39 g of 9-(4-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine obtained in Step 1, 1.72 g of 9'-phenyl-3,3'-bi-9H-carbazole, and 21 mL of mesitylene, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 0.81 g of sodium tert-butoxide, 0.024 g of tris(dibenzylideneacetone)dipalladium(0) (abbreviation: Pd.sub.2(dba).sub.3), and 0.034 g of 2-dicyclohexylphosphino-2',6'-dimethoxybiphenyl (abbreviation: S-Phos) were added thereto, and then stirring was performed at 150° C. for 6 hours for reaction.

(290) After a predetermined time elapsed, the reaction solution was subjected to extraction with toluene. The solid obtained by concentrating the solution of the extract was purified by silica gel column chromatography using toluene as a developing solvent, and then recrystallized with toluene three times, whereby a target substance was obtained (a yellow solid, 1.84 g, in a yield of 62%).

(291) By a train sublimation method, 1.81 g of the obtained yellow solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.7 Pa at 380° C. while the argon gas flowed at a flow rate of 18 mL/min. After the purification by sublimation, 1.35 g of a target yellow solid was obtained in a yield of 75%. The synthesis scheme of Step 2 is shown in Formula (h-2) below.

(292) ##STR00046##

(293) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the yellow solid obtained in Step 2 are shown below.

(294) .sup.1H-NMR. δ (CD.sub.2Cl.sub.2): 7.32-7.39 (m, 2H), 7.44-7.56 (m, 5H), 7.61 (d, 1H), 7.64-7.69 (m, 6H), 7.83-7.91 (m, 6H), 8.11 (d, 1H), 8.17 (d, 1H), 8.28 (d, 2H), 8.49-8.53 (m, 4H), 9.18 (d, 1H), 9.40 (s, 1H).

Reference Synthesis Example 9

(295) A synthesis method of 9-[4-(9'-phenyl-2,3'-bi-9H-carbazol-9-yl)phenyl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9pPCCzPNfpr-02), which is the organic compound represented by Structural Formula (108) in Embodiment 1, is described. The structure of 9pPCCzPNfpr-02 is shown below.

(296) ##STR00047##

(297) Into a three-neck flask were put 1.76 g of 9-(4-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 1 in Reference Synthesis Example 8, 2.22 g of 9'-phenyl-2,3'-bi-9H-carbazole, and 27 mL of mesitylene, and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 1.09 g of sodium tert-butoxide, 0.031 g of tris(dibenzylideneacetone)dipalladium(0) (abbreviation: Pd.sub.2(dba).sub.3), and 0.045 g of 2-dicyclohexylphosphino-2',6'-dimethoxybiphenyl (abbreviation: S-Phos) were added thereto, and then stirring was performed at 150° C. for 6 hours for reaction.

(298) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration and the residue was washed with water and ethanol. The obtained solid was purified by silica gel column chromatography using toluene as a developing solvent, and then recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (a yellow solid, 1.95 g, in a yield of 52%).

(299) By a train sublimation method, 1.94 g of the obtained yellow solid was purified by sublimation. The conditions of the purification by sublimation were such that the solid was heated under a pressure of 2.7 Pa at 380° C. while the argon gas flowed at a flow rate of 18 mL/min. After the purification by sublimation, 1.62 g of a target yellow solid was obtained in a yield of 84%. The synthesis scheme is shown in Formula (i-1) below.

(300) ##STR00048##

(301) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the

yellow solid obtained above are shown below.

(302) .sup.1H-NMR. δ (CD.sub.2Cl.sub.2): 7.28-7.31 (m, 1H), 7.36 (t, 1H), 7.40-7.44 (m, 2H), 7.46-7.51 (m, 3H), 7.57-7.69 (m, 6H), 7.74 (d, 1H), 8.78 (d, 1H), 7.84 (t, 1H), 7.81-7.88 (m, 4H), 8.10 (d, 1H), 8.16 (d, 1H), 8.22 (d, 2H), 8.28 (d, 1H), 8.46 (s, 1H), 8.50 (d, 2H), 9.17 (d, 1H), 9.38 (s, 1H).

Reference Synthesis Example 10

(303) A synthesis method of 9-[3'-(6-phenylbenzo[b]naphtho[1,2-d]furan-8-yl)biphenyl-3-yl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mBnfBPNfpr), which is the organic compound represented by Structural Formula (109) in Embodiment 1, is described. The structure of 9mBnfBPNfpr is shown below.

(304) ##STR00049##

(305) Into a three-neck flask were put 1.28 g of 9-(3-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 1 in Reference Synthesis Example 3, 2.26 g of 3-(6-phenylbenzo[b]naphtho[1,2-d]furan-8-yl)phenylboronic acid pinacol ester, 2.53 g of tripotassium phosphate, 0.89 g of tert-butyl alcohol, and 32 mL of diethylene glycol dimethyl ether (abbreviation: diglyme), and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 8.8 mg of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 28 mg of di(1-adamantyl)-n-butylphosphine (abbreviation: CataCXium A) were added thereto, and then stirring was performed at 140° C. for 8.5 hours for reaction.

(306) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was purified by silica gel column chromatography using toluene as a developing solvent, and then recrystallized with toluene, whereby a target substance was obtained (a yellow solid, 0.66 g, in a yield of 25%).

(307) The synthesis scheme is shown in Formula (j-1) below.

(308) ##STR00050##

(309) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the yellow solid obtained above are shown below.

(310) .sup.1H-NMR. δ (CD.sub.2Cl.sub.2): 7.24-7.28 (m, 3H), 7.61-7.72 (m, 5H), 7.78-7.87 (m, 6H), 7.98-8.00 (m, 3H), 8.08 (d, 1H), 8.11-8.15 (m, 3H), 8.25 (d, 1H), 8.48 (s, 1H), 8.51-8.53 (m, 2H), 8.75 (d, 1H), 9.15 (d, 1H), 9.32 (s, 1H).

Reference Synthesis Example 11

(311) A synthesis method of 9-[3'-(6-phenyldibenzothiophen-4-yl)biphenyl-3-yl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr-02), which is the organic compound represented by Structural Formula (110) in Embodiment 1, is described. The structure of 9mDBtBPNfpr-02 is shown below.

(312) ##STR00051##

(313) Into a three-neck flask were put 1.19 g of 9-(3-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 1 in Reference Synthesis Example 3, 1.97 g of 3-(6-phenyldibenzothiophen-4-yl)phenylboronic acid pinacol ester, 2.29 g of tripotassium phosphate, 0.82 g of tert-butyl alcohol, and 29 mL of diethylene glycol dimethyl ether (abbreviation: diglyme), and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 16 mg of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 52 mg of di(1-adamantyl)-n-butylphosphine (abbreviation: CataCXium A) were added thereto, and then stirring was performed at 140° C. for 15 hours for reaction.

(314) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was purified by silica gel column chromatography using toluene as a developing solvent, and then recrystallized with toluene, whereby a target substance was obtained (a yellowish white solid, 1.17 g, in a yield of

52%). The synthesis scheme is shown in Formula (k-1) below.

(315) ##STR00052##

(316) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the yellowish white solid obtained above are shown below.

(317) .sup.1H-NMR. δ (CD.sub.2Cl.sub.2): 7.39 (t, 1H), 7.47-7.51 (m, 3H), 7.58-7.67 (m, 6H), 7.73 (d, 2H), 7.78-7.85 (m, 5H), 8.02 (s, 1H), 8.06 (d, 1H), 8.10 (d, 1H), 8.18 (d, 1H), 8.23 (t, 2H), 8.49 (s, 1H), 9.17 (d, 1H), 9.30 (s, 1H).

Reference Synthesis Example 12

(318) A synthesis method of 9-{3-[6-(9,9-dimethylfluoren-2-yl)dibenzothiophen-4-yl]phenyl}naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mFDBtPNfpr), which is the organic compound represented by Structural Formula (111) in Embodiment 1, is described. The structure of 9mFDBtPNfpr is shown below.

(319) ##STR00053##

(320) Into a three-neck flask were put 1.01 g of 9-(3-chlorophenyl)naphtho[1',2':4,5]furo[2,3-b]pyrazine whose synthesis method is described in Step 1 in Reference Synthesis Example 3, 1.46 g of 3-[6-(9,9-dimethylfluoren-2-yl)dibenzothiophen-4-yl]phenylboronic acid, 1.89 g of tripotassium phosphate, 0.67 g of tert-butyl alcohol, and 24 mL of diethylene glycol dimethyl ether (abbreviation: diglyme), and the air in the flask was replaced with nitrogen. The mixture in the flask was degassed by being stirred under reduced pressure, 27 mg of palladium(II) acetate (abbreviation: Pd(OAc).sub.2) and 88 mg of di(1-adamantyl)-n-butylphosphine (abbreviation: CataCXium A) were added thereto, and then stirring was performed at 140° C. for 30 hours for reaction.

(321) After a predetermined time elapsed, the obtained suspension was subjected to suction filtration, followed by washing with water and ethanol. The obtained solid was purified by silica gel column chromatography using toluene as a developing solvent, and then recrystallized with a mixed solvent of toluene and hexane, whereby a target substance was obtained (a yellowish white solid, 0.75 g, in a yield of 37%). The synthesis scheme is shown in (1-1) below.

(322) ##STR00054##

(323) Results of analysis by nuclear magnetic resonance (.sup.1H-NMR) spectroscopy of the yellowish white solid obtained above are shown below.

(324) .sup.1H-NMR. δ (CD.sub.2Cl.sub.2): 1.47 (s, 6H), 7.27-7.32 (m, 2H), 7.38 (d, 1H), 7.61-7.76 (m, 8H), 7.79-7.85 (m, 4H), 7.89 (d, 1H), 8.08 (d, 1H), 8.13 (d, 1H), 8.24-8.31 (m, 3H), 8.59 (s, 1H), 9.14 (d, 1H), 9.31 (s, 1H).

Reference Synthesis Example 13

(325) A synthesis method of 11-(3-naphtho[1',2':4,5]furo[2,3-b]pyrazin-9-yl-phenyl)-12-phenylindolo[2,3-a]carbazole (abbreviation: 9mIcz(II)PNfpr), which is the organic compound represented by Structural Formula (112) in Embodiment 1, is described. The structure of 9mIcz(II)PNfpr is shown below.

(326) ##STR00055##

(327) A synthesis method of 9mIcz(II)PNfpr is shown by a synthesis scheme in Formula (m-1) below.

(328) ##STR00056##

Reference Synthesis Example 14

(329) A synthesis method of 3-naphtho[1',2':4,5]furo[2,3-b]pyrazin-9-yl-N,N-diphenylbenzenamine (abbreviation: 9mTPANfpr), which is the organic compound represented by Structural Formula (113) in Embodiment 1, is described. The structure of 9mTPANfpr is shown below.

(330) ##STR00057##

(331) A synthesis method of 9mTPANfpr is shown by a synthesis scheme in Formula (n-1) below.

(332) ##STR00058##

Reference Synthesis Example 15

(333) A synthesis method of 10-[4-(9'-phenyl-3,3'-bi-9H-carbazol-9-yl)phenyl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 10mPCCzPNfpr), which is the organic compound represented by Structural Formula (114) in Embodiment 1, is described. The structure of 10mPCCzPNfpr is shown below.

(334) ##STR00059##

(335) A synthesis method of 10mPCCzPNfpr is shown by synthesis schemes in Formula (o-1) to Formula (o-4) below.

(336) ##STR00060## ##STR00061##

Reference Synthesis Example 16

(337) A synthesis method of 11-[(3'-(dibenzothiophen-4-yl)biphenyl-3-yl]phenanthro[9',10':4,5]furo[2,3-b]pyrazine (abbreviation: 11mDBtBPPnfpr), which is the organic compound represented by Structural Formula (115) in Embodiment 1, is described. The structure of 11mDBtBPPnfpr is shown below.

(338) ##STR00062##

(339) A synthesis method of 11mDBtBPPnfpr is shown by synthesis schemes in Formula (p-1) to Formula (p-7) below.

(340) ##STR00063## ##STR00064## ##STR00065##

Reference Example 17

(341) A synthesis method of 10-[3-(9'-phenyl-3,3'-bi-9H-carbazol-9-yl)phenyl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 10pPCCzPNfpr), which is the organic compound represented by Structural Formula (116) in Embodiment 1, is described. The structure of 10pPCCzPNfpr is shown below.

(342) ##STR00066##

(343) A synthesis method of 10pPCCzPNfpr is shown by synthesis schemes in Formula (q-1) to Formula (q-4) below.

(344) ##STR00067## ##STR00068##

Reference Synthesis Example 18

(345) A synthesis method of 9-[3-(7H-dibenzo[c,g]carbazol-7-yl)phenyl]naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mcgDBCzPNfpr), which is the organic compound represented by Structural Formula (117) in Embodiment 1, is described. The structure of 9mcgDBCzPNfpr is shown below.

(346) ##STR00069##

(347) A synthesis method of 9mcgDBCzPNfpr is shown by synthesis schemes in Formula (r-1) to Formula (r-4) below.

(348) ##STR00070## ##STR00071##

Reference Synthesis Example 19

(349) A synthesis method of 9-{3'-[6-(biphenyl-3-yl)dibenzothiophen-4-yl]biphenyl-3-yl}naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr-03), which is the organic compound represented by Structural Formula (118) in Embodiment 1, is described. The structure of 9mDBtBPNfpr-03 is shown below.

(350) ##STR00072##

(351) A synthesis method of 9mDBtBPNfpr-03 is shown by synthesis schemes in Formula (s-1) to Formula (s-4) below.

(352) ##STR00073## ##STR00074##

Reference Synthesis Example 20

(353) A synthesis method of 9-{3'-[6-(biphenyl-4-yl)dibenzothiophen-4-yl]biphenyl-3-yl}naphtho[1',2':4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr-04), which is the organic compound represented by Structural Formula (119) in Embodiment 1, is described. The structure of 9mDBtBPNfpr-04 is shown below.

(354) ##STR00075##

(355) A synthesis method of 9mDBtBPNfpr-04 is shown by synthesis schemes in Formula (t-1) to Formula (t-4) below.

(356) ##STR00076## ##STR00077##

Reference Synthesis Example 21

(357) A synthesis method of 11-[3'-(6-phenyldibenzothiophen-4-yl)biphenyl-3-yl]phenanthro[9',10':4,5]furo[2,3-b]pyrazine (abbreviation: 11mDBtBPPnfpr-02), which is the organic compound represented by Structural Formula (120) in Embodiment 1, is described. The structure of 11mDBtBPPnfpr-02 is shown below.

(358) ##STR00078##

(359) A synthesis method of 11mDBtBPPnfpr-02 is shown by synthesis schemes in Formula (u-1) to Formula (u-7) below.

(360) ##STR00079## ##STR00080## ##STR00081##

REFERENCE NUMERAL

(361) **101**: first electrode, **102**: second electrode, **103**: EL layer, **103a**, **103b**: EL layer, **104**: charge generation layer, **111**, **111a**, **111b**: hole-injection layer, **112**, **112a**, **112b**: hole-transport layer, **113**, **113a**, **113b**: light-emitting layer, **114**, **114a**, **114b**: electron-transport layer, **115**, **115a**, **115b**: electron-injection layer, **200R**, **200G**, **200B**: optical path length, **201**: first substrate, **202**: transistor (FET), **203R**, **203G**, **203B**, **203W**: light-emitting device, **204**: EL layer, **205**: second substrate, **206R**, **206G**, **206B**: color filter, **206R'**, **206G'**, **206B'**: color filter, **207**: first electrode, **208**: second electrode, **209**: black layer (black matrix), **210R**, **210G**: conductive layer, **301**: first substrate, **302**: pixel portion, **303**: driver circuit portion (source line driver circuit), **304a**, **304b**: driver circuit portion (gate line driver circuit), **305**: sealant, **306**: second substrate, **307**: lead wiring, **308**: FPC, **309**: FET, **310**: FET, **311**: FET, **312**: FET, **313**: first electrode, **314**: insulator, **315**: EL layer, **316**: second electrode, **317**: light-emitting device, **318**: space, **900**: substrate, **901**: first electrode, **902**: EL layer, **903**: second electrode, **911**: hole-injection layer, **912**: hole-transport layer, **913**: light-emitting layer, **914**: electron-transport layer, **915**: electron-injection layer, **4000**: lighting device, **4001**: substrate, **4002**: light-emitting device, **4003**: substrate, **4004**: first electrode, **4005**: EL layer, **4006**: second electrode, **4007**: electrode, **4008**: electrode, **4009**: auxiliary wiring, **4010**: insulating layer, **4011**: sealing substrate, **4012**: sealant, **4013**: desiccant, **4015**: diffusion plate, **4200**: lighting device, **4201**: substrate, **4202**: light-emitting device, **4204**: first electrode, **4205**: EL layer, **4206**: second electrode, **4207**: electrode, **4208**: electrode, **4209**: auxiliary wiring, **4210**: insulating layer, **4211**: sealing substrate, **4212**: sealant, **4213**: barrier film, **4214**: planarization film, **4215**: diffusion plate, **5101**: light, **5102**: wheel, **5103**: door, **5104**: display portion, **5105**: handle, **5106**: shifter, **5107**: seat, **5108**: inner rearview mirror, **5109**: windshield, **7000**: housing, **7001**: display portion, **7002**: second display portion, **7003**: speaker, **7004**: LED lamp, **7005**: control key, **7006**: connection terminal, **7007**: sensor, **7008**: microphone, **7009**: switch, **7010**: infrared port, **7011**: recording medium reading portion, **7012**: support, **7013**: earphone, **7014**: antenna, **7015**: shutter button, **7016**: image receiving portion, **7018**: stand, **7020**: camera, **7021**: external connection portion, **7022**, **7023**: operation button, **7024**: connection terminal, **7025**: band,, **7026**: microphone, **7027**: icon indicating time, **7028**: another icon, **7029**: sensor, **7030**: speaker, **7052**, **7053**, **7054**: information, **9310**: portable information terminal, **9311**: display portion, **9312**: display region, **9313**: hinge, **9315**: housing

(362) This application is based on Japanese Patent Application Serial No. 2018-147134 filed with Japan Patent Office on Aug. 3, 2018, the entire contents of which are hereby incorporated herein by reference.

Claims

1. A light-emitting device comprising an electroluminescent layer between an anode and a cathode, wherein the electroluminescent layer comprises: a first layer; a second layer between the first layer and the cathode and in contact with the first layer; a third layer between the second layer and the cathode and in contact with the second layer; and a fourth layer between the cathode and the third layer and in contact with the third layer and the cathode, wherein the second layer has a largest thickness in the electroluminescent layer, wherein the second layer comprises a light-emitting substance, a first organic compound and a second organic compound, wherein the first layer comprises the first organic compound, wherein the third layer comprises a third organic compound, wherein the fourth layer comprises a compound comprising an alkali metal, wherein the third organic compound is different from the first organic compound and the second organic compound, wherein the second organic compound has a structure in which an aromatic ring is fused to a furan ring of a furodiazine skeleton, and wherein the second layer is a light-emitting layer.
2. The light-emitting device according to claim 1, wherein the compound comprising an alkali metal is an organic compound.
3. The light-emitting device according to claim 1, wherein the fourth layer further comprises the first organic compound.
4. The light-emitting device according to claim 1, wherein the aromatic ring in the structure included in the second organic compound is a naphthalene ring.
5. The light-emitting device according to claim 4, wherein the furodiazine skeleton in the second organic compound is a furopirazine skeleton or a furopyrimidine skeleton.
6. The light-emitting device according to claim 1, wherein the first organic compound is represented by any one of Formulae (G1), (G2) and (G3), ##STR00082## wherein Ar.sup.1 represents a substituted or unsubstituted aromatic ring, wherein each of R.sup.1 to R.sup.6 independently represents hydrogen or a group having 1 to 100 carbon atoms in total, and wherein each of at least one of R.sup.1 and R.sup.2, at least one of R.sup.3 and R.sup.4, and at least one of R.sup.5 and R.sup.6 has a hole-transport skeleton.
7. The light-emitting device according to claim 1, wherein the first organic compound is represented by any one Formulae (G1), (G2) and (G3), ##STR00083## wherein Ar.sup.1 represents any one of substituted or unsubstituted benzene, substituted or unsubstituted naphthalene, substituted or unsubstituted phenanthrene, and substituted or unsubstituted chrysene, wherein each of R.sup.1 to R.sup.6 independently represents hydrogen or a group having 1 to 100 carbon atoms in total, and each of at least one of R.sup.1 and R.sup.2, at least one of R.sup.3 and R.sup.4, and at least one of R.sup.5 and R.sup.6 has a hole-transport skeleton.
8. The light-emitting device according to claim 6, wherein Ar.sup.1 represents any one of Formulae (t2), (t3) and (t4), ##STR00084## wherein each of R.sup.11 to R.sup.36 independently represents any one of hydrogen, a substituted or unsubstituted alkyl group having 1 to 6 carbon atoms, a substituted or unsubstituted monocyclic saturated hydrocarbon group having 3 to 7 carbon atoms, a substituted or unsubstituted aromatic hydrocarbon group having 6 to 30 carbon atoms, and a substituted or unsubstituted heteroaromatic hydrocarbon group having 3 to 12 carbon atoms, and wherein * represents a portion bonded to the furan ring in Formulae (G1) to (G3).
9. The light-emitting device according to claim 6, wherein the group having 1 to 100 carbon atoms in each of R.sup.1 to R.sup.6 of Formulae (G1) to (G3), represents one of or a combination of an alkyl group having 1 to 6 carbon atoms, a substituted or unsubstituted monocyclic saturated hydrocarbon group having 5 to 7 carbon atoms, a substituted or unsubstituted polycyclic saturated hydrocarbon group having 7 to 10 carbon atoms, a substituted or unsubstituted aromatic hydrocarbon group having 6 to 13 carbon atoms, and a substituted or unsubstituted heteroaromatic hydrocarbon group having 3 to 12 carbon atoms.
10. The light-emitting device according to claim 6, wherein the group having 1 to 100 carbon atoms in each of R.sup.1 to R.sup.6 of Formulae (G1) to (G3), has any one of a pyrrole ring

- structure, a furan ring structure and a thiophene ring structure through one of a substituted or unsubstituted phenylene group and a substituted or unsubstituted biphenylene group.
11. The light-emitting device according to claim 6, wherein the group having 1 to 100 carbon atoms in each of R.sup.1 to R.sup.6 of Formulae (G1) to (G3), has a structure represented by any one of Formulae (Ht-1) to (Ht-26) through one of a substituted or unsubstituted phenylene group and a substituted or unsubstituted biphenylene group, ##STR00085## ##STR00086## wherein Q represents oxygen or sulfur, wherein each of R.sup.100 to R.sup.169 represents 1 to 4 substituents and independently represents any one of hydrogen, an alkyl group having 1 to 6 carbon atoms, and a substituted or unsubstituted aromatic hydrocarbon group having 6 to 13 carbon atoms, and wherein Ar.sup.1 represents a substituted or unsubstituted aromatic hydrocarbon group having 6 to 13 carbon atoms.
12. The light-emitting device according to claim 1, wherein the light-emitting substance is a phosphorescent material.
13. The light-emitting device according to claim 1, wherein the first organic compound is a carbazole derivative.
14. The light-emitting device according to claim 13, wherein the carbazole derivative is a bicarbazole derivative.
15. The light-emitting device according to claim 1, wherein the first layer is a hole-transport layer.
16. A light-emitting apparatus comprising: the light-emitting device according to claim 1; and a flexible printed circuit.
17. An electronic device comprising: the light-emitting apparatus according to claim 16; and at least one of a microphone, a camera, an operation button, an external connection portion and a speaker.
18. A lighting device comprising: the light-emitting device according to claim 1; and at least one of a housing and a cover.
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